

Skyseam Build Spec

Oct 8, 2026 · @sb

1. Read this first: how Claude Code should use this document

This is the build specification for **Skyseam**, a Minecraft 1.21.1 NeoForge mod. It is the Design Bible with one change: every item now says who builds it. Sections 1 and 2 tell Claude Code how to read the rest, and they win over anything below them. Section 23 holds the locked answers to every open question, so there is nothing to ask the author before starting.

Reading order

1. Sections 1 and 2, then section 23.
2. Sections 3 to 15 and 17 for the design. Skim 3 and 4 for the story and read 5 to 15 for the systems.
3. Section 19 for the milestone order. Build in that order.
4. Sections 20 to 22 alongside each milestone: the asset list, the audio plan and the manual list.
5. Section 16 is a prompt for a different project (Fealty), so do not run it. Section 18 is a backlog, so build nothing from it unless the author names the item.

The five tags

Every item that has to be built carries one of these tags, always written in code style.

Tag	What it means	Claude Code should	Claude Code must not
[CC]	Claude Code builds it and can check it itself	Build it fully. It is done when the build, the data generator and its GameTest or headless server check pass	Leave a TODO where the feature should be

Tag	What it means	Claude Code should	Claude Code must not
[FIRST PASS]	Claude Code builds a complete, working first version. Only the author can judge how it looks, sounds or feels	Build the real thing, wired in and easy to swap. Look at what it can (generated PNGs, renders). Report it as "built, needs your eyes or ears"	Call it final, or say that it looks or sounds good
[SUPPORT ONLY]	Claude Code cannot make the content, and the content is optional ("add it later if we need it")	Build the hooks, config, registrations and empty slots, plus a how-to-add guide, so the author can drop the content in with no code change. The game must run cleanly with the slot empty	Invent substitute content or ship fake files
[MANUAL]	Claude Code cannot make it, and the final mod will contain it. The author's words: "I'll add this later manually"	Build a working stub, add one line to <code>TODO-MANUAL.md</code> , and keep going	Block on it, guess at it, or skip it silently
[OTHER REPO]	Built in a different project by a different session (the Fealty features)	Write only the Skyseam side, against the documented API, inside one package (<code>`compat/fealty`</code>)	Edit the other project

An item with no tag: treat art, sound and structures as [FIRST PASS] and everything else as [CC]. Text with no tag is design background.

Standing rules

- 1. Do not wait on the author.** The open questions are answered in section 23. Where this spec is silent, pick the simplest reading, record it in `docs/DECISIONS.md` and continue. Stop and ask only for a real blocker: a dependency that cannot be downloaded, a license problem, or a choice that cannot be undone.
- 2. Never block on [SUPPORT ONLY] or [MANUAL].** Build the plumbing or the stub, record it, move on.

3. **Never fake what you cannot make.** No placeholder "music", no empty files with real-looking names, no text that pretends to be final. A stub is labeled in `TODO-MANUAL.md` and in `docs/STATUS.md`.
4. **Everything is swappable without code.** Textures, models, animations, sounds, music, structures and lang files sit at the paths in section 20 and are found by name. Replacing a file with a file of the same name must work with no code change.
5. **Everything tunable is data or config.** Each number in this document that is marked config or tune goes into a config file or a data pack JSON, with the document's value as the default.
6. **Placeholders first.** Every entity, block and item works as a plain placeholder before any art exists, so quests and fights can be tested early.
7. **Fealty is required, the rest stay optional.** Fealty is a required dependency, declared as required in `neoforge.mods.toml`, and Skyseam compiles against its API version 1.3.0. FTB Quests and KubeJS are soft dependencies: Skyseam starts and plays without them, so guard every use with a loaded-mod check and keep their classes out of the main code path.
8. **Nothing copied.** Use Create, Create Aeronautics (the Simulated project) and Sable through their public APIs only. Never copy their code, models, textures or sounds.
9. **One bridge for outside ids.** Every Sable, Simulated and Create class, id and method is touched in one place (`SableBridge` plus an `ExternalIds` class), with pinned versions. This document read those names from source at an earlier version, so confirm each one against the pinned jar before relying on it.
10. **Server decides, client shows.** Gameplay logic runs on the server. The client plays visuals and sounds from packets.
11. **Look at what you make.** Generated textures, model renders and structure previews are PNG files you can open and view. Do that before reporting a `[FIRST PASS]` item. You still cannot hear audio or see the running game.
12. **Report honestly.** At each milestone write `docs/STATUS.md` : what was built, how it was verified, what only the author can check, and what is still a stub. Commit once per milestone with the milestone name in the message.

Files to create

File	Purpose
<code>docs/ENVIRONMENT.md</code>	Written first. The answers to the environment checks in section 2
<code>docs/DECISIONS.md</code>	One line for each choice made where this spec was silent
<code>docs/STATUS.md</code>	Per milestone: built, how it was verified, what needs the author's eyes or ears, what is stubbed
<code>docs/ASSET-PATHS.md</code>	Every asset kind, its folder, its naming rule and how to replace it (section 20)
<code>docs/ADDING-MUSIC.md</code>	The exact steps to fill each music slot (section 21)
<code>TODO-MANUAL.md</code>	One line per <code>[MANUAL]</code> item and per <code>[SUPPORT ONLY]</code> slot: what it is, which file goes where, and what the game does until then. Seed it from section 22
<code>reference/</code>	The four reference images from section 3, put there by the author. If they are missing, build from the written descriptions in section 14 and say so in <code>docs/DECISIONS.md</code>

2. Who builds what

This section is the source of truth for responsibility. If a later section disagrees with the table below, the table wins. The tags are explained in section 1.

Step zero: check the environment

Before writing mod code, create `docs/ENVIRONMENT.md` with the answers to these checks. Several tags depend on them, and the right fallback differs between a cloud sandbox and the author's own machine.

Check	Why it matters
Java 21 and the Gradle wrapper run, and Gradle can reach the NeoForge, Create, Sable, GeckoLib and Modrinth Maven repositories	Without this nothing builds. Report it and stop
<code>runServer</code> , <code>runData</code> and <code>runGameTestServer</code> run headless	They are the main automatic checks
Sable's native physics library loads on a headless server	Decides whether the M0 spike runs as a GameTest or needs the author
Which Blockbench MCP is connected: the desktop plugin, the headless <code>.bbmodel</code> server, or neither	Decides the model pipeline
A GPU and Node 23.6 or newer, for the headless renderer	Without them Claude Code cannot preview a model
Python 3 with Pillow and numpy, an OGG writer (ffmpeg with libvorbis, or the soundfile package) and an NBT writer (the nbtlib package)	Textures, synthesized sounds and structures
A display, real or virtual, for <code>runClient</code>	Lets Claude Code launch the client and read its log for model or sound errors

Responsibility table

Area	Tag	How	Checked by
Mod code: registries, entities, AI, blocks, items, menus, packets, systems, config	<code>[CC]</code>	NeoForge Java, with GeckoLib 4 for animated entities and block entities	<code>./gradlew build</code> , GameTests, headless server boot

Area	Tag	How	Checked by
Data: recipes, loot tables, advancements, tags, dimension, biomes, noise settings, physics pack, lang keys, the <code>sounds.json</code> structure	[CC]	<code>runData</code> or hand-written JSON. Version 1.21 folders are singular (<code>recipe</code> , <code>advancement</code> , <code>loot_table</code> , <code>structure</code>)	<code>runData</code> , server boot, a <code>GameTest</code> per recipe and advancement
Ship crossing between dimensions	[CC] , gated by the M0 spike	Route A or B in section 6, with the fallback in section 23	A <code>GameTest</code> if Sable loads headless, otherwise a dev command the author runs
Seam, Echo Sounder, sky and beam rendering code, particle types	[CC] code, [FIRST PASS] look	Custom renderers, additive quads, particle providers	It compiles and runs. The look is the author's call
Screens and menus	[CC] logic, [FIRST PASS] art	Screens, menus and payloads, with panels and sprites drawn by script	Menu <code>GameTests</code> . The author checks layout in game
Entity and block-entity models, animations, glow masks	[FIRST PASS]	Blockbench MCP if connected, otherwise generated geo JSON. Sources live in <code>art/models/</code>	MCP validators when present, the client log, renders if there is a GPU, and the author's eyes
Textures for blocks, items, entities, particles, sky and GUI	[FIRST PASS]	Pillow and numpy generators in <code>tools/textures/</code> , or the MCP paint tools	Claude Code views the PNGs. The author's eyes

Area	Tag	How	Checked by
Structures (all 15)	[FIRST PASS]	Generator scripts in <code>tools/structures/</code> write <code>.nbt</code> files. Hero builds are parametric (towers, domes, arches, rings) and use the pictures in <code>reference/</code>	A GameTest places each file and checks size and block counts. Claude Code views isometric previews
Sound effects	[FIRST PASS]	Vanilla stand-ins through <code>"type": "event"</code> entries, plus synthesized mono <code>.ogg</code> files for the hero sounds (<code>tools/audio/</code>)	A script validates files, subtitles and <code>sounds.json</code> , and measures length, peak and loop seams. Claude Code cannot hear them
Music	[SUPPORT ONLY]	Music manager, empty slots and a how-to guide (section 21)	The game runs cleanly with every slot empty
Tuning, balance, difficulty	[FIRST PASS]	This document's numbers as defaults	The author's playtests
The eight Fealty features	[OTHER REPO]	The prompt in section 16, run in the Fealty project	Fealty's own GameTests
Skyseam side of Fealty: rumour data, conversations, favor objectives, the Stargazer	[CC]	Built against Fealty 1.3.0 in M10, after the Fealty project finishes the section 16 work	GameTests with Fealty on the classpath
Advancements, custom stats, tags	[CC]	Data generation	<code>runData</code> , and one GameTest per advancement chain
FTB Quests chapters, quests and rewards	[MANUAL]	The author builds them from the ids in section 17	Not applicable
Final sound design for the hero moments	[MANUAL]	Replace the synthesized files by name (section 22)	The author's ears

Area	Tag	How	Checked by
Final art for the hero pieces and the hero structures	[MANUAL]	Replace the first-pass files by name (section 22)	The author's eyes
Real-hardware testing: performance, shader packs, multiplayer, dedicated server	[MANUAL]	Claude Code boots a dedicated server and writes the tests it can	The author
Mod icon, page art, trailer, license check, publishing	[MANUAL]	A plain placeholder icon only	The author

What Claude Code cannot do

- Hear audio. It can measure a file (length, peak level, loop seam) and view a spectrogram, but it cannot judge taste.
- See the running game. It can view any PNG, so generated textures, model renders and structure previews are checkable, but the game window is not.
- Compose music or record sound.
- Judge feel, difficulty, framerate or shader-pack behavior.
- Run a real multiplayer session, or publish the mod.

Tools that help (all optional)

Blockbench MCP (jasonjgardner/blockbench-mcp-plugin). It comes as two servers, and either or both can be connected.


```
# Desktop plugin over HTTP. Blockbench desktop must be running with the
plugin loaded
# (File > Plugins > Load Plugin from URL:
https://jasonjgardner.github.io/blockbench-mcp-plugin/mcp.js)
claude mcp add blockbench --transport http http://localhost:3000/bb-mcp
```

```
# Headless server over stdio. No Blockbench needed. --root limits file access
to that folder
claude mcp add bbmodel --transport stdio -- npx -y
github:jasonjgardner/blockbench-mcp-plugin --root /path/to/art/models
```

- The headless server edits `.bbmodel` files (`bbmodel_create` , `bbmodel_edit` , `bbmodel_add_texture`), checks them (`bbmodel_validate` , which includes GeckoLib rules, and `bbmodel_validate_animations`), exports Bedrock geometry (`bbmodel_export_bedrock_geometry`) and renders PNG previews (`bbmodel_render` , `bbmodel_contact_sheet` , GPU needed). Its documentation lists no animation export, so Claude Code writes each `.animation.json` itself from the model's keyframes (it is plain JSON) and checks the clips with `bbmodel_validate_animations` .
- The desktop plugin adds painting and animation tools and screenshots of the 3D view, which is the only way to see a model without a GPU render. Its `risky_eval` tool runs any JavaScript it is sent, so leave it switched off in Blockbench settings unless it is needed.
- Treat both as experimental. If a call fails, fall back to generated geo JSON and note it in `docs/ENVIRONMENT.md` .

An audio-generation MCP or API, only if the author has one. Nothing in this document depends on it. Section 21 explains how the audio is covered without it.

3. The pitch

Build tags. Background only. Nothing in this section is built directly. The things it names are built in the sections that follow.

Skyseam sends an airship through a crack in the sky into a small, bounded pocket world of floating islands, mirror seas and living light, and makes you earn the way back out.

A villager you have won over whispers about a scar in the sky. Five strange favors later you craft the **Harmonic Aperture**, mount it on a Create Aeronautics ship and fly to the marked spot. The sky unpicks itself like a seam and your whole ship slips into **the Halcyon**, an archipelago about 3 km across, lit by a single hovering lantern-orb instead of a sun. Inside

are helper creatures that produce things you truly need, three Stay Guardian bosses, a central puzzle spire, and the parts for two tools: the **Pneumatic Coupler** (a craftable copy of the creative physics staff) and the **Echo Sounder** (a sonar tool, and the only way to find the hidden exit seams).

Design pillars

- **Flight is the verb.** Arrival, travel, boss arenas, docking and loot are all built around airships.
- **Calm, beautiful, slightly wrong.** The world looks serene. Danger arrives as events and set pieces, not constant mob spam.
- **Every helper is a lever.** No cosmetic-only creatures: each passive mob produces something the player needs.
- **Earn the exit.** You cannot leave until you have built the Echo Sounder and found a hidden seam.
- **Quest-native.** Every milestone is a discrete advancement, stat or item tag, so FTB Quests needs no custom code.
- **Small enough to know by heart.** A bounded world you can learn, with replay value from Tides and events.
- **No armor.** Rewards are tools, utilities, consumables and blocks.

Working names

All names are working names. Keep them unless the author changes them in section 23. Put every player-visible name in the lang file so renaming is one pass.

Thing	Working name	Alternates
Mod and namespace	Skyseam (skyseam)	Halcyon Rift, Opaline
The portal	The Seam	The Rent, the Scar
Pocket dimension	The Halcyon	The Lull
Entry block (the final craft)	Harmonic Aperture	Seam Ripper, Seam Key
Staff copy	Pneumatic Coupler	the author's call
Sonar tool	Echo Sounder	Resonance Sounder
Map from the villager	Skychart	Seam Chart

Thing	Working name	Alternates
Quest villager	Wystan Starling, the Stargazer	any name
Ancient builders	The Wrights (Skyfolk)	none

How the four reference images are used

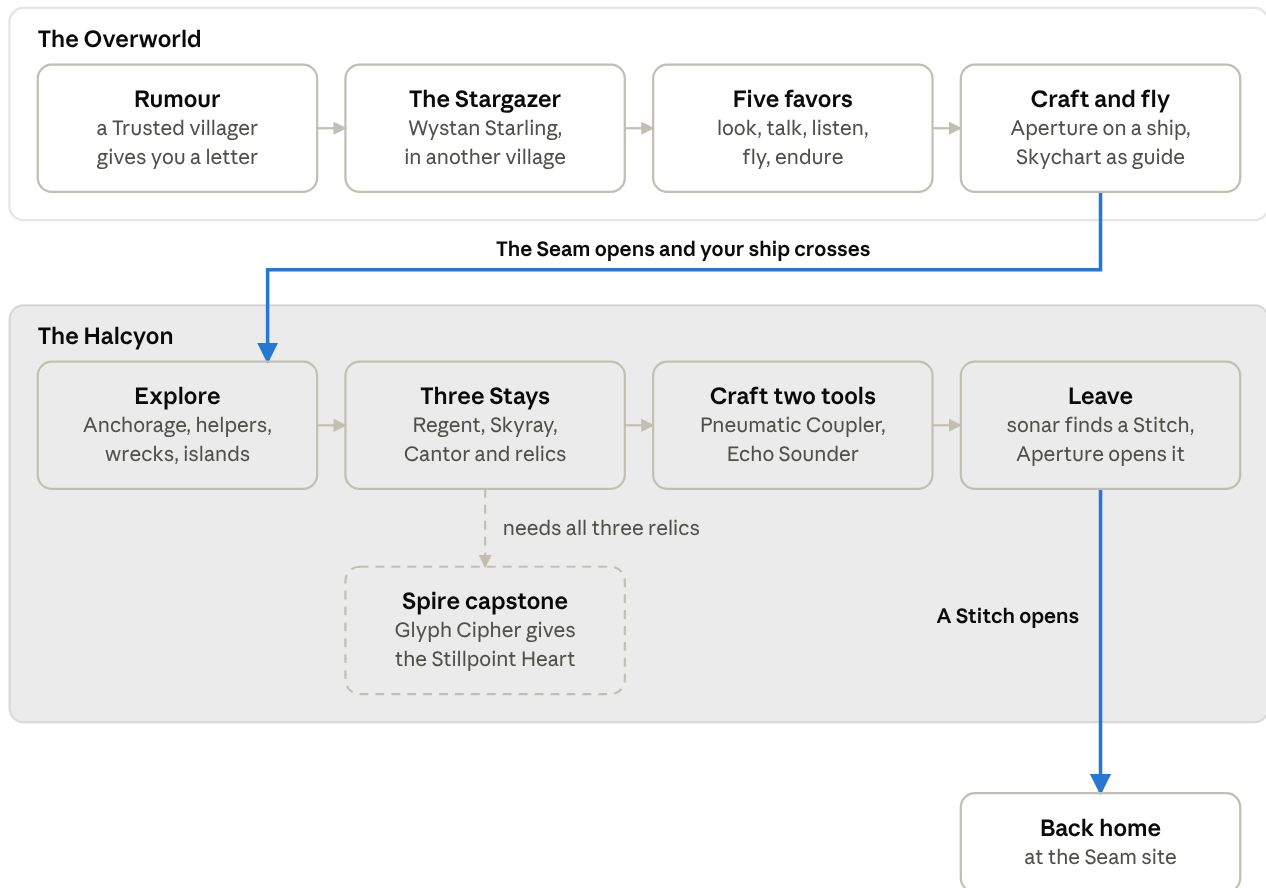
The four pictures are style references. Claude Code should open them from `reference/` before building the Seam look, the sky and the structures (sections 6, 7 and 14). If the files are named differently, match them by the descriptions below.

Image	Becomes
1. Rift from Minecraft Dungeons	The Seam’s look: ragged voxel-edged fissure, white glowing outline, pastel prismatic interior, dust streaming off one edge
2. Arch, glowing orb, petal field, mirror lake	The Halcyon’s mood board: Lantern-Sun, Petalwash Meadows, Mirror Sea, and the Standing Arch gate to the first boss
3. Split floating monolith over a cliff gate	The Sundered Obelisk, the landmark you see first on arrival (its gap rhymes with the Seam)
4. Domed overgrown tower, teal and gold	The Conservatory Spire, the central puzzle tower with one themed floor per relic

The locked decisions are in section 23.

The journey at a glance

Five favors get you in; a sonar gets you out



journey at a glance · 2 worlds, 2 doors, 10 steps

The Seam is the door in and a Stitch is the door out. The dashed Spire is a capstone you can reach any time after the three Stays, and it is not needed to leave.

1. **Rumour.** A Trusted villager mentions the scar in the sky and hands you the Starlit Letter.
2. **The Stargazer.** Wystan Starling lives in a different village, marked by a Brass Telescope on a rooftop, and opens up to the letter.
3. **Five favors.** Look, talk, listen, fly and endure. Each favor rewards one component.
4. **Closing talk.** He asks why you want to go, then hands over the Keystone and the Skychart.
5. **Craft.** The Harmonic Aperture, from the five components, the Keystone and two Create parts.

6. **Fly.** Bolt the Aperture to an Aeronautics ship and follow the Skychart to the marked site, at least 30 blocks above the ground.
7. **The Seam opens.** About six seconds of charge, hairline, crack and snapping golden threads, then the ship crosses once it overlaps the opening.
8. **Arrive.** The Anchorage on the Sundered Obelisk island: a Mooring Post, Wrightling stalls and the first Almanac pages.
9. **Explore.** Helpers give Moonsilk, Antler Glass and Whale Song Shells, and the wrecks hold logbook pages.
10. **Three Stays.** The Prism Regent, Tempest Skyray and Hollow Cantor in any order, each dropping a relic.
11. **Craft two tools.** The Skyray's Pneumatic Heart builds the Pneumatic Coupler, and the Cantor's Resonant Tine builds the Echo Sounder.
12. **Find a Stitch.** A ping draws a dotted outline of one of the four hidden exits for 20 seconds.
13. **Leave.** Charge the Aperture beside the Stitch and you return to the Seam site in the Overworld.
14. **Capstone.** Put the three relics into the Glyph Cipher at the Spire's summit for the Stillpoint Heart.

4. Lore

Build tags. Background for Claude Code. Every piece of in-game text that comes from this lore (the twelve logbook pages, Almanac entries, Wrightling lines, the carved motto) is `[FIRST PASS]`. Write it in `assets/skyseam/lang/en_us.json` in the tone below, and the author will reword whatever they do not like.

The Halcyon is a shelter the Wrights stitched into the sky, and the Seam is the one stitch they left loose.

The Wrights were airship builders who watched a storm they called the Unmooring tear the sky open. They wove a pocket of permanent calm out of the torn fabric, sailed their fleet inside, and sewed the opening shut. Villagers still tell it as a nursery story: the sky has a scar, and the old stargazers keep the needle.

Three machines called the **Stays** hold the pocket's calm: the Mirror Stay (light), the Wind Stay (air) and the Echo Stay (sound). Over the centuries their guardians stopped recognising visitors and began treating every ship as the storm.

Why you cannot just leave. The Wrights hid the exits so the storm could never trace them. A hidden stitch can only be heard with an Echo Sounder, and its parts are guarded by the Stays. The Wright motto is carved at the Anchorage: *no one leaves who cannot listen*.

The wrecks. A few Wright ships never reached the Anchorage. Their twelve logbook pages teach the glyph script that the Spire's final puzzle is written in, so exploring the wrecks is how knowledge, not just loot, becomes progress.

Tone

- Wistful and lonely, never grim. The Halcyon is abandoned, not evil.
- Story arrives through logbooks, Wrightling dialogue and the Almanac, never cutscenes.
- Beauty first: the first thirty seconds after arrival should make the player stop flying and look.

5. Part 1: Before the Rift

Build tags. [CC] the chain logic, the five favor objectives, the clue generator, the rumour roll, the items below, and the Star Dial and Bell Sequence puzzles. [FIRST PASS] every model, texture, GUI sprite and sound, and the wording of the dialogue and rumour lines. [OTHER REPO] the Fealty features this chain stands on (sections 15 and 16). Fealty is a required dependency, so this chain is built in M10, after the Fealty project has finished the section 16 work.

The Overworld half is a six-step chain built on Fealty's existing rumour, quest-giver and dialogue systems: hear a rumour, meet the Stargazer, finish five favors that each yield one component, receive the Keystone and the Skychart, then craft the Harmonic Aperture.

1. **Rumour.** Once a player is Trusted in a village, ordinary villagers have a rare chance to mention the scar in the sky.
2. **The Stargazer.** Wystan Starling lives in a different village, marked by a Brass Telescope on a rooftop. He only opens up to a player carrying the Starlit Letter from the rumour.
3. **Five favors.** Each uses a different verb (look, talk, listen, fly, endure) and rewards one component.
4. **Closing talk.** He asks why you want to go. Your answer picks a small cosmetic title and unlocks the Keystone and the Skychart.
5. **Craft** the Harmonic Aperture from the five components, the Keystone and two Create parts.
6. **Mount and fly.** Bolt it to an Aeronautics ship and follow the Skychart.

The rumour roll

- Eligible: Trusted tier or higher (reuses the start tier of Fealty's rare villager chain).
- Chance: 3 percent per conversation, once per villager per day. Each miss adds 1 percent (cap 12). Guaranteed after 30 eligible conversations. All values are config.
- Three stages, each delivered by a different villager so players meet several: **Whisper** (the scar exists), **Lead** (a Survey Map to the Stargazer's village), **Introduction** (the Starlit Letter, a sealed letter item).
- Sample Whisper lines: "My flock won't graze under that patch of sky." / "The bells stopped ringing the day the light split." / "Ask a stargazer, not a farmer."
- Write at least 30 Whisper lines, grouped by villager profession, in the same voice as the samples.

The five favors

Favor	Verb	What the player does	Component	Objective id
The Lost Survey	Look	Follow riddle clues from the Survey Notes to the Hollow Cairn and align the three-ring Star Dial on a clear night	Star-Glass Lens	skyseam:survey (extends explore)
Three Hands, One Seal	Talk	Win a sealed accord from the elders of three Trusted villages, each with its own dialogue challenge (settle a feud by checking facts, deliver a crafted item, return a runaway animal)	Triple-Wax Seal	skyseam:three_seals (builds on courier and Fealty letters)

Favor	Verb	What the player does	Component	Objective id
The Silent Chapel	Listen	Find the Silent Chapel, recover three bell clappers, then play the sequence carved on the wall	Tuned Clapper	skyseam:chapel_bells
The Ring Run	Fly	Fly a ship carrying the Flight Logbook through 12 Wind Rings over the village inside four minutes	Windwheel Gear	skyseam:ring_run
The Stormglass Night	Endure	Build a Skyspike lightning tower, hold the empty vial on it through a thunderstorm and beat the Stormlings it draws	Stormglass Vial	skyseam:stormglass

Register each objective id as a Fealty quest type.

Clues for the Cairn and the Chapel are generated per world by locating the nearest structure, the same way Fealty's chain maps are made, so no world is unwinnable. A lost component can be re-earned from the Stargazer for a fee, with the puzzle skipped.

Keystone and Skychart

- **Skywright's Keystone** is the one ingredient that cannot be crafted. It is the Stargazer's own heirloom.
- **Skychart** is a map item marking the Seam site. With an Aperture mounted it also shows the live gauge from the next section.

Harmonic Aperture recipe

Star-Glass Lens	Precision Mechanism	Triple-Wax Seal
Brass Casing	Skywright's Keystone	Brass Casing
Tuned Clapper	Windwheel Gear	Stormglass Vial

The two Create parts are `create:precision_mechanism` and `create:brass_casing` (confirm the ids against the pinned Create version). The block is soulbound to the crafter, and teammates can use it if the pack runs FTB Teams.

6. The Seam (portal)

Build tags. `[CC]` the Seam entity, trigger rules, charge logic, gauge, force-loading, ship transfer, mending and the debug commands. `[FIRST PASS]` the look of the reveal (renderer, particles, interior sky, camera shake) and every sound in the beat table. The ship transfer is gated by the M0 spike in section 19.

The Seam opens only while a qualifying ship is flying near the marked site. It appears at the ship's altitude, takes the whole ship through, and mends when nobody is left near it.

Trigger rules

Rule	Value (all config)
Site	One (x, z) per world, 600 to 900 blocks from the Stargazer's village, saved in world data and drawn on the Skychart
Entry radius	48 blocks horizontal from the site
Ship	A Sable sub-level containing a Harmonic Aperture, with a player piloting it
Flying	Ship speed above 2 blocks per second and no ground contact
Altitude	Lowest ship block at least 30 blocks above the ground at the site (motion-blocking height ignoring leaves at that x, z)
Dimension	Overworld by default

Rule	Value (all config)
Charge	5 seconds with every rule met. Dropping a rule pauses and drains the gauge
Spawn point	(site x, ship y, site z), facing the ship's approach bearing
Size	Seam width and height = ship bounds x 1.5, clamped to 16 to 64 blocks, so large ships fit
Hold radius	128 blocks. The Seam mends 5 seconds after the last qualifying ship leaves, or after 120 seconds regardless

The reveal, beat by beat

Beat	Time	Player sees and hears
1. Listening	during charge	Aperture core spins up with a rising hum. A HUD gauge shows distance and three ticks (flying, altitude, radius). Heat shimmer at the site
2. Hairline	0 to 1.5 s	One thin white vertical line, a glass "tink", drifting motes
3. Crack	1.5 to 3.5 s	Voxel-edged cracks branch outward in steps, like screenshot 1. Cloth-tearing sound with a glass chime. Clouds part in a ring
4. Threads snap	3 to 5 s	Golden threads along the edges snap one at a time, each with a harp pluck and a spark, and each snap widens the opening a notch
5. Open	5 to 6 s	Pastel prismatic interior fades in (pink, orange, teal), dust streams off one edge, the white outline pulses, soft god-rays, light camera shake
6. Stable	loop	Low hum, edge shimmer, a slow ring pulse every 6 s, and a gentle pull on ships within 60 blocks
7. Crossing	on overlap	When the ship's bounds overlap the opening by 60 percent: pearl-white flash, then the Halcyon arrival title

Beat	Time	Player sees and hears
8. Mending	on leave	Threads re-stitch bottom to top, the crack narrows, a closing chime, faint scar particles for 60 s during which it cannot reopen

Everyone within 256 blocks sees the animation, not just the pilot.

How it should be built

- The Seam is an **entity** with a custom renderer, not a block, because it hangs in open air at any height.
- The interior is a pre-baked Halcyon sky texture with parallax, not a true portal render. It is cheap and stays compatible with shader packs.
- Edges are additive line strips and translucent quads. Everything else is particles and sound.

Moving the ship

1. Freeze the ship and riders, then snapshot blocks, block entities, entities inside the bounds, and velocity.
2. Choose a clear Arrival Lane in the Halcyon (64 by 64 blocks of open air at Y 240).
3. Move it. **Route A:** Sable's own save and load path (`SubLevelSerializer.toData` then `fullyLoad` into the Halcyon level). **Route B:** copy blocks across and rebuild with `SubLevelAssemblyHelper.assembleBlocks`.
4. Restore velocity at 50 percent, place riders at the same relative positions, and mend the Overworld Seam.

Both routes exist in Sable's public code, but neither is proven for cross-dimension moves. The first prototype task is a spike that moves a test ship between two dimensions. Default size cap: 64 by 64 by 64 blocks. If the spike shows that neither route works, use the fallback in section 23 (decision 2) and record the result in `docs/DECISIONS.md`.

Edge cases

- Only ships with an Aperture cross. Anyone standing on the ship rides along. People on foot elsewhere are left behind.
- Riders who log out are saved with the snapshot and reappear at the Arrival Lane.
- Chunks around the site stay force-loaded while the Seam is open.

- Losing the pilot, or the Aperture, mid-charge cancels the charge.
- A creative-only command opens a Seam anywhere for testing.
- Operator-only debug commands live under `/skyseam` : open and mend a Seam, force a Tide, give the Almanac, and teleport to landmarks. Claude Code needs them for GameTests and the author needs them for filming.

7. The Halcyon (pocket dimension)

Build tags. `[CC]` the dimension and dimension type, terrain generation, biomes, fixed landmark positions, the Veil, Rebound, the Lantern-Sun cycle logic, the physics data pack and the flora blocks. `[FIRST PASS]` the look: sky and fog colors, the Lantern-Sun art, the cloud texture, biome palettes and the ambience loops. `[SUPPORT ONLY]` music (the music line under Flora and sound).

The Halcyon is a bounded archipelago about 3 km across, with a mirror sea below and a lantern-orb above, built so that flying is the best way to see it.

Shape and size limit

Property	Value (starting point, tune in playtests)
Radius	1,500 blocks around the Spindle at the center, so the world is 3,000 by 3,000 blocks
Heights	Mirror Sea at Y 96, islands from Y 140 to 320, sky lid at Y 420
The Veil	A soft boundary at the radius with a fade. Over the outer 200 blocks the fog thickens, colors wash out to pearl white and sound muffles. Ships are deflected like light off glass and players feel a prismatic push. No invisible wall and no death: anyone who crosses the radius is carried gently back, and anyone who jumps or falls off the edge drops into the Mirror Sea and triggers Rebound
The floor	The Mirror Sea never kills: falling into it triggers Rebound (see Core systems)
Arrival	Arrival Lane on the south-west rim at Y 240, with the Sundered Obelisk visible ahead

Sky and light

- There is no sun or moon. The **Lantern-Sun**, a hovering orb like the one in screenshot 2, circles the Spindle at Y 360 on a 40 minute cycle.
- Its position sets four phases: Dawn-Glass, Noon-Bloom, Dusk-Prism and the Hush (the orb dims to an ember, stars and aurora appear). Mobs and several puzzles read the phase.
- The skybox is a pastel gradient with slow cloud banks and a faint ring on the horizon, the inner wall of the pocket. Fog is tinted per biome.

Physics pack

Sable lets a data pack set per-dimension physics in

`data/skyseam/dimension_physics/halcyon.json`. Proposed start: gravity -9 (Overworld is -11), base pressure 1.2 with the curve staying flat to Y 320, universal drag 0.07. The air

feels thicker and floatier, so ships behave a little differently here and players notice without being told. Confirm the folder name and the fields against the pinned Sable version.

Biomes

Biome	Look	Terrain	Purpose
Petalwash Meadows	Purple petal carpets, glass-stem flowers, still pools (screenshot 2)	Wide, gentle floating plateaus	Safest home of the helper creatures
Mirror Shoals	Shallow reflective water, pale rock outcrops	Low reefs over the Mirror Sea	The Standing Arch and the Prism Regent
Cirrus Reefs	Cloud-coral islands, soft cloud blocks, drifting petals along wind lanes	Airy, stacked reefs	Whale roosts and the Gale Ring
Underbloom	Inverted gardens: hanging roots, lantern vines, glow moss	The undersides of islands	Hidden entrances, Cogwren nests. You must fly under islands
The Hush	Dim blue-violet, stars and aurora, muted sound	Deep hollows and tall spires	Hollow Listeners, the Belfry and the Hollow Cantor
The Spindle	A column of rising light ringed by orbiting islands	Concentric ring islands	The Conservatory Spire (screenshot 4)
Wreckfields	Graveyards of Wright ships snagged on drifting reefs	Scattered wreck islands	Salvage, logbook pages, brass

Terrain generation

- Floating islands come from 3D density noise with three size classes (small 20 to 40 blocks, medium 60 to 120, large 150 to 300). Biomes form rough concentric rings with noise at the borders.

- Key structures sit at **fixed positions** (Obelisk, Spire, three arenas) so quests and maps can name them. Everything between is generated. The four exit seams pick from nine candidate sites using the world seed.
- Use the simplest implementation that gives a bounded size, fixed landmark positions and the three island size classes. At 3,000 by 3,000 blocks, generate lazily per chunk and keep island density in config so the larger area stays playable. Data-driven noise settings with density functions, or a custom `ChunkGenerator`, are both fine.
- Dimension type: tall enough for the sky lid (for example `min_y` 0 and `height` 512). The sky is drawn by a custom `DimensionSpecialEffects` (NeoForge's `RegisterDimensionSpecialEffectsEvent`; confirm the name in the pinned version).

Flora and sound

- **Glassreed** chimes when brushed, and pitch follows wind speed. **Lumen Lily** glows and bounces players. **Cloudmoss** grows on cloud blocks. **Prismite** crystal clusters are mined for Prism Dust.
- Music is `[SUPPORT ONLY]`. The intended sound is sparse piano and glass harmonica, one layer per Tide. Claude Code cannot make music, so it builds the music slots described in section 21 and leaves them empty. The Hush stays nearly silent on purpose even when music exists, because sound is a gameplay channel there.
- Ambience is `[FIRST PASS]`: one synthesized loop per Tide and one for the Hush (section 21).

8. Core systems

Build tags. `[CC]` all nine systems. `[FIRST PASS]` the visual and audio side: beam and ring rendering, Tide weather visuals, Almanac art and every sound.

Nine reusable rules carry the whole dimension, so every creature, puzzle and boss is built from the same few ideas and players learn the Halcyon once.

System	The rule	Used by
Tides	Weather events on top of the day phase. Calm is default. Every 25 minutes or so one Tide runs for 6 to 10 minutes: Prismfall (glass-shard rain, Prism Dust to gather, Shardlings spawn), Gale (crosswinds, tailwinds on lanes, Gale Harriers), Bloom (flowers open, helpers produce double, Pollen Bloats drift in), Stillness (rare: wind stops, sound deepens). A Tide Vane block and the Almanac show the next one	Creatures, bosses, gathering
Wind Lanes	Visible ribbons of petals and light motes between landmarks. A ship inside a lane is nudged along it (a force if Sable allows, else a velocity nudge each tick). Direction flips with Tides	Travel, the Skyray fight, Create windmill puzzle
Refraction	A light beam leaves the Lantern-Sun, hits Prism Pylons you rotate in 15 degree steps, splits into red, green and blue at a Splitter and recombines. Max 12 bounces, 64 blocks per leg	Prism Regent, Spire dome
Echo	Sound is a channel. Pings, sprinting, explosions and bell strikes create noise events with a radius. Listeners and the Hollow Cantor react. A ping reveals hidden things as outlines for 20 seconds	Echo Sounder, Hush, Belfry

System	The rule	Used by
Rebound	Falling into the Mirror Sea bounces you 24 blocks up with 8 seconds of Slow Falling, then Weakness for 10 seconds, at most once per 30 seconds. A ship hitting the sea is sprung back up without damage. No void deaths, and none at the border either	The floor of the world
Moorings	A Mooring Post docks a ship (it holds still within 6 blocks), stores a chest-sized cargo hold, and sets your respawn inside the Halcyon until you first leave through an exit	Anchorage, wrecks, camps
Almanac	A codex item with five tabs: Chart (discoveries on a map), Bestiary (fills by observing), Glyphs, Tides, Favors. Every entry is also an advancement	FTB Quests, hints
Salvage and Glyphs	Twelve logbook pages are hidden in the Wreckfields. Each teaches one glyph pair. The Spire's last puzzle is a six-glyph phrase. A Wren House near a wreck lets Cogwrens deliver one page per Tide cycle	Spire, helpers
Wrightling rep	Wrightlings are tiny artisans at the Anchorage and Moorings with their own Fealty faction <code>skyseam:wrights</code> . Gifts of Moonsilk, brass and flowers raise rep. Tiers unlock Lost and Found (replaces a lost key item), Tide forecast, repair-kit trades and hints	Services, Fealty tie-in

Fealty's API says data packs can add static factions, so the Wrightling faction needs no code change in Fealty.

9. Creatures

Build tags. [CC] entity classes, AI goals, spawn rules, loot, products, rep effects and dialogue hooks. [FIRST PASS] every model, animation, texture and sound. The rig list is in section 20, and creature sounds start as vanilla stand-ins (section 21).

The Halcyon has four helper species whose products are required for progress, a few ambient animals, and six hostile types that arrive mostly through Tides and the Hush. No creature wears or drops armor.

Helpers that matter

Creature	Look	Behavior	What it contributes
Lanternmoth	Palm-sized glowing moth, four soft wings	Gathers around lit lanterns, dims and rests in the Hush	Shear its cocoon for Moonsilk (3 per cocoon, regrows each Tide). Moonsilk becomes the Sounder's membrane and Resonance Cells. Also a living light source
Wayfarer Whale	14-block sky whale, slow and gentle	Follows wind lanes and sings. At each Tide change it drifts toward the nearest exit seam you have not found	Right-click a singing whale with a Hollow Shell to record a Whale Song Shell (10 minute cooldown per whale), a Sounder part. Its drift is a soft hint when you are lost. Harming it costs Wrightling rep
Cogwren	Chicken-sized bird with brass feathers	Nests in wrecks. A Wren House block nearby makes it fetch one Logbook Page per Tide cycle until all 12 are found	Unlocks the glyphs for the Spire without a fixed route. Also picks up dropped items near a Mooring

Creature	Look	Behavior	What it contributes
Bloomstag	Slender deer with crystal antlers	Grazes petal carpet, sheds antlers during Bloom	Antler Glass (2 per shed, never harmed), the rod of the Pneumatic Coupler
Wrightling	Knee-high artisan in a brass apron	Lives at the Anchorage and Moorings, speaks through the dialogue system	Trades, Lost and Found, hints, Tide forecast (see Wrightling rep)

Ambient and neutral

Creature	Notes
Glasskipper	School of ray-like fish skimming the Mirror Sea. They scatter when a Hollow Listener is near, a free warning for watchful players
Updraft Gull	Flock that follows ships. Feed Glass Berries and they circle wind lanes, making them easy to see
Brasshorn Ram	Grazes ledges and headbutts anyone who walks up from the front. Knockback only

Hostile

Creature	Where and when	Behavior	Drop
Shardling	Prismfall, Mirror Shoals	Knee-high glass imps in swarms of 6 to 10. Throw shards, shatter when hit hard	Prism Dust
Hollow Listener	The Hush	Tall, blind, bell-eared. Hunts by noise only: sprinting, pings, explosions. Sneaking past is safe	Muted Bell (consumable, silences you for 30 s)

Creature	Where and when	Behavior	Drop
Thread Spinner	Underbloom, Wreckfields	Spider-like, strings golden thread tripwires between islands. Threads slow ships and players. Shears cut them	Golden Thread
Gale Harrier	Gale Tide, Cirrus Reefs	Raptors that dive at ship decks and knock passengers about. Reward deck design	Gale Plume
Stray Reflection	Mirror Shoals	A humanoid of mirror that walks on the sea and copies your movements a second late. Hits you if you stand still	Mirror Shard
Pollen Bloat	Bloom Tide	A drifting jelly that bursts into Levitation for 6 seconds. A hazard near ships, a tool near ledges	none

Overworld only: the **Stormling**, a small static-charged wisp that appears in thunderstorms near a Skyspike during the Stormglass Night favor.

10. Bosses

Build tags. [CC] the three fights: phases, arena logic, multipart hit boxes, drops, per-player relics and the Mooring Post at each entrance. [FIRST PASS] models, animations, textures and fight sounds. [SUPPORT ONLY] the boss themes: three music slots, each with an optional per-phase stem hook (section 21). Boarding the Skyray is the hardest part, and its fallback is in section 19.

Three Stay Guardians, one per Stay, each teaching a core system (light, wind, sound). The Tempest Skyray is the fight that yields the Pneumatic Coupler parts. Recommended order is Regent, Skyray, Cantor, but any order works, and the Spire needs all three relics. Tune each fight to about 7 minutes with iron-tier gear and expect packs to retune it.

1. The Prism Regent (Mirror Stay)

- **Arena:** the Hall of Reflections, entered through the Standing Arch from screenshot 2, whose mirror door previews the hall. A disc of shallow water over a mirror floor, ringed by six Prism Pylons, with the Lantern-Sun overhead.
- **Look:** a tall slender crystal figure with a glass crown and eight orbiting shard ribbons.
- **Phase 1, Reflections:** it splits into four copies. Only the real one casts a reflection in the floor. Decoys pop when hit.
- **Phase 2, Refraction:** players rotate the Pylons to route the sun beam onto the Regent's core while it casts Overexposure. A hit stuns it for 8 seconds at triple damage.
- **Phase 3, Shatter:** the floor breaks into nine hovering platforms and Stray Reflections join. Falling into the sea only costs time because of Rebound.
- **Drops:** Prism Core (Spire relic), 3 to 5 Mirror Shards.

2. The Tempest Skyray (Wind Stay), Coupler boss

- **Arena:** the Gale Ring, eight small islands around a central cyclone column, open sky on every side.
- **Look:** a 48-block manta made of cloud-stone plates, with six glowing air sacs along its flanks.
- **Phase 1, Circling:** it glides in loops and throws charged cloud cells that arc to metal on ships. Players ground the arcs with lightning rods or pop the cells, so deck design matters.
- **Phase 2, Cyclone:** the central storm grows. Crosswinds push ships and the Skyray's wake pulls them. Ride the wind lanes or moor to the ring islands.
- **Phase 3, Low pass:** it skims a Perch island for 15 seconds so players can jump onto its back and pop all six air sacs while they are inflated, exposing the heart.
- **Drops: Pneumatic Heart** (Coupler core), 2 to 4 Gale Plumes, Gale Core (Spire relic).
- **Highlights:** boarding a flying colossus, and thunder rods bolted to your own deck.

3. The Hollow Cantor (Echo Stay)

- **Arena:** the Belfry, a vast hollow island in the Hush with a ring of giant Bell Pillars and almost no light.
- **Look:** a hooded choir-master of dark glass with twelve small bells orbiting it. It is invisible until pinged.

- **Phase 1, Call and response:** it plays a four-note phrase on the pillars. Replaying it correctly stuns and reveals it for 10 seconds.
- **Phase 2, Choir:** it calls Hollow Listeners. Noise draws them, so fight quietly, sneak-attack, or use Muted Bells.
- **Phase 3, Crescendo:** phrases grow to six notes and the arena rings with false echoes, so listen for pitch, not just position.
- **Drops:** Echo Core (Spire relic), **Resonant Tine** (Sunder part).
- **Gear for the fight:** a Tuning Fork (short-range ping) given by the Wrightlings at the Belfry door.

Shared rules

- Each arena has a Mooring Post at its entrance, a respawn point, and a safe return route, so a loss never traps the player.
- Kills are tracked per player. Co-op fights share the drop roll but every participant gets their own relic.
- Each fight has a music slot with an optional per-phase stem hook (section 21). With no music file the fight is silent apart from its sound effects.

11. Items, blocks, recipes

Build tags. [CC] every item, block, recipe, loot entry and tag. [FIRST PASS] all item and block art (section 20). The Pneumatic Coupler needs no art under Option A.

The mod adds about thirty items, a set of working blocks and a building set, and no armor. Every craft uses something a creature or boss gives, so the Halcyon's life matters to the player.

Key tools and blocks

Name	Kind	Obtained	What it does
Harmonic Aperture	Block	Crafted from the five favor components, the Keystone and two Create parts	Opens the Seam when mounted on a flying ship. Has a gauge GUI
Skychart	Item	Stargazer, after the fifth favor	Marks the Seam site, shows the entry gauge

Name	Kind	Obtained	What it does
Almanac	Item	Stargazer, at the closing talk	Five-tab codex (Chart, Bestiary, Glyphs, Tides, Favors)
Pneumatic Coupler	Item	Crafted in the Halcyon	An exact copy of Create Aeronautics' creative physics staff
Echo Sounder	Item	Crafted in the Halcyon	Sonar pulse and beam that reveals hidden exit seams
Tuning Fork	Item	Wrightlings at the Belfry door	Short-range ping for the Cantor fight
Prism Spyglass	Item	Spyglass plus Mirror Shard	Shows wind lane direction and names creatures at range
Mooring Post	Block	2 brass ingots, 3 sticks, 1 Moonsilk	Docks ships, stores cargo, sets respawn
Prism Pylon and Splitter	Blocks	Mirror Shard plus brass	The light-routing pieces
Tide Vane	Block	Brass plus Prism Dust	Shows the next Tide
Wren House	Block	4 planks, 1 brass nugget, 1 Golden Thread	Lets Cogwrens deliver logbook pages
Halcyon Lantern	Item	Lumen Lily, brass, glass	Portable light that draws Lanternmoths

Pneumatic Coupler

Mirror Shard		Gale Plume		Mirror Shard
Antler Glass		Pneumatic Heart		Antler Glass
Precision Mechanism		Brass Casing		Precision Mechanism

The Heart and Plume come from the Tempest Skyray, Antler Glass from Bloomstags, and the shards from the Regent or Stray Reflections.

Making it an exact copy. The Create Aeronautics source, read when this was written, shows that the staff is `simulated:creative_physics_staff`, class `PhysicsStaffItem`, epic rarity, stack size 1, range 128 blocks. Its handlers only check `instanceof PhysicsStaffItem`, and no creative-mode check was found, so it works in survival once held. Confirm the id, the class and the range against the pinned version before relying on any of it. Two ways to ship the copy:

- **A. Recipe only (recommended, locked in section 23).** The recipe's result is the original item with its name changed to "Pneumatic Coupler". No item code, identical behavior and look. In 1.21.1 a recipe result can carry data components. Use `minecraft:item_name` so the name shows as plain text, or `minecraft:custom_name` with italics off if the staff's tooltip depends on its default name. Add a GameTest that crafts it and asserts that the result is the Simulated staff item with the new name.
- **B. Subclass (not chosen).** `PneumaticCouplerItem` extends `PhysicsStaffItem` under `skyseam:pneumatic_coupler`. Still recognized by every handler, but needs its own model, because Aeronautics' art assets are all rights reserved and cannot be copied.

Echo Sounder and Resonance Cell

Mirror Shard	Moonsilk	Mirror Shard
Brass Ingot	Resonant Tine	Brass Ingot
Whale Song Shell	Precision Mechanism	Moonsilk

Resonance Cell (shapeless): 1 Moonsilk, 1 Prism Dust, 1 copper ingot. One cell powers a Pulse, three power a Beam.

Materials

Moonsilk, Prism Dust, Mirror Shard, Gale Plume, Antler Glass, Golden Thread, Hollow Shell, Whale Song Shell, Pneumatic Heart, Resonant Tine, Muted Bell, Logbook Page, and Glass Berries (food: restores hunger and gives 5 seconds of Glowing).

Building set

Stillstone stone, bricks, slabs, stairs, walls, pillars and glass for the Wright architecture, plus Petal Carpet, Cloud blocks, Glassreed and Lumen Lily for dressing islands.

Armor rule

No armor, no trinket slots, no Curios dependency. Survivability comes from systems (Rebound, Mooring respawns, Muted Bells), not gear.

12. Leaving the Halcyon

Build tags. [CC] the Stitches, the exit charge, the Echo Sounder logic, noise events, the radar data and the soft-lock nets. [FIRST PASS] the look of the ping ring and beam, the radar art and every sound.

Four hidden exit seams lead home, and the Echo Sounder is the only reliable way to find them, so the player is stuck until they have built it.

The exits (Stitches)

- Four Stitches, chosen at world creation from nine candidate sites in different biomes and heights. At least one sits on an island summit reachable on foot, one in the Underbloom and one deep in the Hush.
- **Dormant** means invisible and intangible. The only tell is a faint shimmer and thread hum within 12 blocks, which attentive players can learn to notice.
- A ping from the Sounder draws a dotted outline for 20 seconds and logs the site on the Almanac Chart.
- **Opening one:** a ship with a Harmonic Aperture within 24 blocks runs the same charge as the entry, with the reveal played in reverse (the threads unpick). A player on foot opens a walkable Stitch by carrying the Aperture item within 6 blocks.
- Every Stitch returns you to the Seam site in the Overworld, at your departure altitude or 30 blocks above ground, whichever is higher.

The Echo Sounder

Mode	How it works	Range	Cost
Pulse	Hold 1 second. A ring of pale-blue light expands from you, sweeping terrain and open air	96 blocks	1 Resonance Cell
Beam	Hold 3 seconds while aiming. A narrow cone with ripples along the line	400 blocks	3 Resonance Cells

- **Look and sound:** the ring clips through islands, things it hits glow and chime, and pitch drops with distance. A low thrum plays on release and the echoes return a moment later.
- **Screen:** held in hand it shows a round radar plan view with a compass ring, blips that stay unlabeled until identified, and the last ten pings.
- **Reveals:** dormant seams, wreck islands, large creatures as shapes, Prismite deposits and Mooring Posts. Boss arenas are always visible.
- **The risk:** a ping is a noise event. Hollow Listeners within 128 blocks come to the spot. In the Hush the Cantor's choir stirs. Prismfall and Gale cut range by 30 percent. Stillness doubles range and noise.

The gate: building the Sounder

1. Collect Moonsilk from Lanternmoths.
2. Record a Whale Song Shell from a singing Wayfarer Whale.
3. Gather Mirror Shards from Stray Reflections or the Prism Regent.
4. Defeat the Hollow Cantor for the Resonant Tine, using the Tuning Fork, which also teaches the risk of pinging.
5. Craft it, then refill with Resonance Cells.

That forces at least one boss, two helper species and a trip through three biomes before leaving.

Soft-lock safety nets

- Dying in the Halcyon respawns you at a Mooring Post, never in the Overworld, until you first leave through a Stitch.
- Dropped items stay at the death spot. Wrightling **Lost and Found** replaces lost key items for a fee once Wrightling rep is high enough.
- A lost ship: once per Tide cycle a Wrightling will assemble a basic Wright Skiff at the Anchorage for a fee.
- Whales drift toward unfound exits at each Tide change, a free hint for the stuck.
- A config switch turns all four nets off for hardcore packs, and an admin command ejects a stuck player.

13. Puzzles and GUIs

Build tags. [CC] the logic of all twelve puzzles, the per-world shuffling, menus, payloads and server-side checks. [FIRST PASS] the art and layout of every screen, the puzzle block models and the puzzle sounds. The author checks layout and readability in game.

Twelve puzzles, no two sharing a mechanic, spread across the Overworld chain, the three arenas and the Spire. Answers that depend on the world (angles, orders, glyph phrases) are shuffled per world, so a video guide cannot hand out one solution.

Puzzle catalog

#	Puzzle	Where	Mechanic	Reward
1	Star Dial	Hollow Cairn (Overworld)	Three-ring dial GUI. Rotate rings to match constellation clues from the Survey Notes	Star-Glass Lens
2	Bell Sequence	Silent Chapel (Overworld)	Place three clappers, then strike the bells in the order carved on the wall	Tuned Clapper
3	Ring Run	Sky above the village	Fly through 12 Wind Rings in order within 4 minutes	Windwheel Gear
4	Skyspike	Overworld thunderstorm	Build a lightning tower, keep the vial on it, and repel Stormlings	Stormglass Vial
5	Prism Routing	Hall of Reflections, Spire dome	Rotate Pylons so a colored beam reaches matching receivers	Regent stun, dome opens
6	Flower Clock	Petalwash vault	Flowers open at certain Lantern-Sun phases. Arrange colors to match three phase markers, then wait	Vault cache and Glass Berry seeds

#	Puzzle	Where	Mechanic	Reward
7	Thread Loom	Spire gardens floor	GUI grid. Route golden threads between node pairs without crossing	Lift unlocks
8	Echo Map	The Hush caves	Pulse pings in the dark and follow rising pitch to three hidden bells, struck low to high	Belfry door opens
9	Gravity Switch	Wreck interiors	Float and Sink plates toggle levitation fields so you walk walls and ceilings	Logbook Page
10	Wind Lane Turbine	Spire arcade	Build a Create windmill or propeller engine in a wind lane that meets the lift's gauge	Powers the Spire lift
11	Mirror Maze	Mirror Shoals ruin	Walls are mirrors that lie. The water floor shows the true layout	Mirror Shard cache
12	Glyph Cipher	Spire summit	Three relic sockets and a six-glyph wheel. Translate the phrase using logbook pages	Stillpoint Heart

Screens

Skyseam ships a small UI kit styled to match Fealty's screens (tab bar, scroll list, buttons) so the two mods feel like one set. If the author supplies Fealty's GUI textures in `reference/fealty-gui/`, match them. Otherwise use a neutral pastel-glass style and say so in `docs/DECISIONS.md`.

Screen	Purpose
Dialogue	Reuse Fealty's dialogue box: portrait, subtitle, up to six replies with icons, and grayed replies that explain what they need

Screen	Purpose
Aperture	Status ticks, charge ring, Skychart slot (mockup below)
Seam Gauge HUD	Compact overlay near the hotbar while an Aperture is mounted
Almanac	Chart, Bestiary, Glyphs, Tides, Favors tabs
Star Dial	Three concentric rotating rings over a star field
Thread Loom	Grid of nodes with drag-to-connect thread
Glyph Cipher	Three relic sockets and a rotating glyph wheel
Echo Sounder	Radar plan view (mockup below)
Wrightling services	Trade, Lost and Found, Tide forecast, hints

```
+-----+
| HARMONIC APERTURE          owner: you |
|                               |
|      ( charge ring  62% )      |
| [x] Flying  [x] Altitude  [ ] Site |
|                               |
| Site: 1204 / -388    Distance: 212 |
| Ground at site: Y 71    Need: Y 101+ |
| [ Skychart slot ]           |
+-----+
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    .----- .
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Cells 2/3   Mode: Beam   Range 400
```

14. Structures

Build tags. [FIRST PASS] all fifteen structures: Claude Code writes generators that produce the .nbt files from the reference images and the palette rule. [CC] the world-gen registration, placement rules, loot tables, structure sets, data markers and the GameTests that place every file. The author may later overwrite any file with a hand-built version of the same name ([MANUAL] , optional, section 22).

Fifteen structures, three of them tied directly to the reference images. Palette rule: pale limestone and warm brown brick, moss green, teal and gold accents, violet crystal veins, soft white light. Nothing black, nothing harsh.

Halcyon

Structure	Where	Reference	Holds
Sundered Obelisk	Anchorage, arrival island	Screenshot 3: a floating monolith split in two with a small orb in the gap, mossy, with a cliff gateway at the base	Mooring Post, Wrightling stalls, the carved motto, the first Almanac pages
Conservatory Spire	The Spindle	Screenshot 4: overgrown domed tower, teal and gold dome, balconies, hanging gardens. About 120 blocks tall	Four floors: flooded arcade (Wind Lane Turbine), gardens (Thread Loom), observatory dome (Prism Routing), summit (Glyph Cipher)
Standing Arch	Mirror Shoals	Screenshot 2: a 40-block arched mirror gate standing in water beside pale rocks, a glowing orb nearby	Entrance to the Regent's hall
Hall of Reflections	Beyond the Arch	Disc of shallow water over a mirror floor	Regent arena, six Pylon plinths
Gale Ring	Cirrus Reefs	Eight islands around a cyclone column	Skyray arena, a Mooring Post and a lightning mast on each island

Structure	Where	Reference	Holds
The Belfry	The Hush	A hollow island with a ring of Bell Pillars	Cantor arena, Echo Map puzzle
Wright Wrecks	Wreckfields	Three sizes: 30, 60 and 120 blocks	Logbook pages, brass, the Gravity Switch rooms
Moth Grove	Petalwash Meadows	Lantern trees around a pool	Lanternmoth cocoons
Whale Roost	Cirrus Reefs	A hollow cloud arch with bone-white rings	Where whales rest and sing
Wright Mooring	Scattered	A small dock with one Wrightling	Respawn and storage
Stitch rings	Nine candidate sites	A ring of frayed-stitch stones, easy to miss	The hidden exits

Overworld

Structure	Where	Holds
Hollow Cairn	Mountain biomes	Standing stones, a toppled brass telescope, the Star Dial pedestal
Silent Chapel	Plains and forests	Ruined chapel with three empty bell frames and wall carvings
Brass Telescope	On a rooftop in the Stargazer's village	The marker that tells players they found him
Wind Ring course	Sky above the village	Entities, no blocks, so nothing to generate

How the structures get built

- **Generated by script.** Generators in `tools/structures/` (Python with an NBT writer, or a small Java tool that uses Minecraft's own NBT classes) write `data/skyseam/structure/<name>.nbt`. Generators are deterministic for a given seed and size.

- **Hero builds are parametric.** The Conservatory Spire is a tall tower with a teal and gold dome, balconies and hanging gardens. The Sundered Obelisk is a split monolith with a gap and an orb socket above a cliff gate. The Standing Arch is a 40-block arch standing in shallow water. For each one, open the matching picture in `reference/`, match its silhouette, palette and proportions, and view an isometric preview (`tools/structures/preview.py`) before moving on.
- **Repeatable pieces are the natural fit for generators:** floating islands in three size classes, ring courses, hollow cave islands, wreck hulls and Stitch rings.
- **Props and creatures are not structures.** They are modeled (section 20).
- **Data markers.** Mark every functional spot (puzzle blocks, chests, spawn points, each Stitch center) with a data marker, list the markers in `docs/STRUCTURE-MARKERS.md`, and let code swap each marker for the real block or block entity after placement. A hand-built replacement then only needs the same markers.
- **Placement.** Overworld structures use normal structure generation (a structure set with rare spacing and biome tags). Halcyon landmarks sit at fixed positions chosen by the dimension generator (section 7).
- **Check.** A GameTest places every `.nbt` file in a flat test world, asserts the bounding box and that every marker listed for it exists, and flags blocks outside the palette list.
- **Swap rule.** The author can hand-build any structure in creative, save it with a structure block under the same name and overwrite the file. That is the optional polish step in section 22. No code change should be needed.

15. Fealty integration

Build tags. `[OTHER REPO]` the eight Fealty features listed below, written out as a prompt in section 16. `[CC]` everything on the Skyseam side: the Fealty compat package, the rumour and conversation data files, the five favor objectives and the Stargazer entity. `[FIRST PASS]` all dialogue wording and the models of the Stargazer and the Wrightling.

Fealty already has most of what this needs: a rumour chain, a quest-giver system, server-driven dialogue and a quest-type registry. The plan is to build on those and add eight small, general features to Fealty, not a Skyseam-specific hack. The Fealty source (NeoForge 1.21.1, API 1.2.0) was read from the author's zip.

What Fealty already gives us

Fealty piece	Where it lives	How Skyseam uses it
Rep tiers and API	<code>FealtyApi</code> , <code>RepApi</code> , data-driven tiers	Gate the rumour roll (Trusted or higher), read rep for dialogue conditions
Rare villager chain	<code>QuestChainDefinition</code> , <code>ChainManager</code> , <code>data/fealty/fealty/quest_chains/</code>	The model for the Stargazer chain: rumours, steps, a map, a trial, a final reward
Dialogue	<code>DialogueService</code> , <code>DialogueNode</code> , <code>VillagerDialogue</code> , client <code>DialogueScreen</code>	The conversation UI for villagers, the Stargazer and Wrightlings
Quest objectives	<code>QuestObjective</code> , <code>QuestType</code> , registry <code>fealty:quest_type</code>	Skyseam registers its five favor objectives here
Quest givers	<code>QuestGiver</code> , <code>QuestGivers</code> , <code>KeeperEntity</code>	The Stargazer is a new giver
Letters	<code>SealedLetterItem</code> , mailboxes	The Starlit Letter and the three accords in Three Hands
Factions	Data-driven factions	<code>skyseam:wrights</code> for Wrightling rep
FTB Quests and KubeJS	<code>compat/ftbquests</code> , <code>compat/kubejs</code>	Fealty rep and quest events already reach the author's questbook

One naming clash: Fealty already has a **Keeper** (the hidden hamlet NPC). This design calls ours the **Stargazer** to avoid confusing the two.

What Fealty needs added

1. **Single-giver chains.** Today each step is set by a different villager. Add a chain mode where one named NPC gives every step in order.
2. **A stable API for quest types, chain kinds and quest givers.** The quest-type registry currently lives in the main jar. Promote it to the API jar.

3. **A rumour hook.** An event and data format for rumours with conditions and weights, plus a rare roll on ordinary villager conversations (today only an elder's "rumours" action starts the chain).
4. **Branching dialogue from data.** Nodes, replies with conditions (rep tier, item, flag, advancement, dimension, time) and effects (take or give item, set flag, change rep, start quest, grant advancement).
5. **Per-player flags.** A small `getFlag` and `setFlag` API so a conversation can remember a choice.
6. **Dialogue contributors.** Let another mod add a reply or line to an existing villager's node, for example "Ask about the sky".
7. **Special-role villagers.** A public way to mark one villager as a named role with a title (Fealty already stores roles on villagers).
8. **Events.** `RumourEvent`, `DialogueEvent` and `ChainStageEvent`, so FTB Quests and KubeJS can react.

Section 16 is a ready prompt for the Fealty project. It is not part of this build.

What Claude Code builds on the Skyseam side

1. **Fealty is required.** Declare it as a required dependency in `neoforge.mods.toml` and compile against its API version 1.3.0. Do not write a fallback dialogue runner.
2. **Order of work.** The Fealty project finishes the section 16 prompt first. Until then Claude Code builds everything that does not touch dialogue, rumours or Fealty quest types. Dialogue-dependent features (the Stargazer, Wrightling talk, the rumour roll, the favor objectives) are built in M10.
3. **One package for Fealty.** All Fealty calls live in one package (`compat/fealty`). The Fealty session may name things slightly differently from the prompt, so read the finished `docs/API.md` in the Fealty project first and adapt inside that package only.
4. **Data where Fealty reads it.** Put every conversation, rumour and chain file at `data/skyseam/fealty/conversations/`, `data/skyseam/fealty/rumours/` and `data/skyseam/fealty/quest_chains/`. The formats are in this section and in section 16.
5. **What it registers.** The five favor objectives as Fealty quest types, the static faction file for `skyseam:wrights`, the Stargazer role on a villager in a found village (a Brass Telescope is placed on a rooftop there), and listeners that forward Fealty events to advancements.

Dialogue data format

```
{
  "id": "skyseam:stargazer/first_meeting",
  "speaker": "skyseam:stargazer",
  "start": "greet",
  "nodes": {
    "greet": {
      "text": "skyseam.dialogue.stargazer.greet",
      "replies": [
        { "label": "skyseam.reply.show_letter", "if": { "has_item":
"skyseam:starlit_letter" }, "goto": "letter" },
        { "label": "skyseam.reply.ask_sky", "goto": "sky" },
        { "label": "skyseam.reply.bye", "end": true }
      ]
    },
    "letter": {
      "text": "skyseam.dialogue.stargazer.letter",
      "effects": [ { "take_item": "skyseam:starlit_letter" }, { "set_flag":
"skyseam:met_stargazer" } ],
      "replies": [ { "label": "skyseam.reply.ready", "goto": "favors" } ]
    }
  }
}
```

Example: first meeting with the Stargazer

- **Wystan:** "Mind the stairs. The telescope has right of way."
- Reply A (needs the Starlit Letter): "A villager sent me. I have this." Reply B: "What are you looking at?" Reply C: "Goodbye."
- **A, then Wystan:** "Mara's hand, and Mara's bad spelling. Then she told you. Good. Sit, and listen, because the sky does not stay open long." He keeps the letter and sets the `met_stargazer` flag.
- **B, then Wystan:** "A stitch that has come loose. Come back with a letter and I will show you where." (Reply A stays hidden until the letter exists.)
- Further replies unlock the five favors one at a time, each with a short briefing and a clearly grayed reply that explains what is missing when the player is not ready.

Fealty is a required dependency (decision 4 in section 23).

16. Prompt for the Fealty AI

Build tags. [OTHER_REPO] Claude Code does not run this prompt. The author pastes it into the session that has the Fealty project open. Claude Code only needs to know what it asks for, so that the Skyseam compat package (M10) can use the result. The text inside the box was edited so that Fealty stays standalone and names no other mod.

The author pastes this whole block into the session that has the Fealty project open. It names the real files found in the Fealty zip, asks for eight general features and nothing Skyseam-specific, and ends with tests it must pass. A copy was also sent to the author as a file.

TASK

Extend the Fealty mod with general extension points so data packs and other mods can add story chains, rumours and branching conversations. Fealty must keep working completely on its own: it must not depend on, require or mention any other mod. Every feature below must be general, data-driven where possible, and backwards compatible.

CONTEXT

- Project: Fealty (NeoForge 1.21.1, mod id fealty, package com.selluastar.fealty, API 1.2.0, see docs/API.md).
- Read before coding: chain/QuestChainDefinition.java, chain/ChainManager.java, quest/QuestObjective.java, quest/QuestGiver.java, quest/QuestGivers.java, quest/ElderGiver.java, registry/ModQuestTypes.java, registry/FealtyRegistries.java, registry/ModAttachments.java, dialogue/DialogueService.java, dialogue/DialogueNode.java, dialogue/VillagerDialogue.java, dialogue/GiverDialogue.java, data/FealtyDataManager.java, data/CodecDataLoader.java, village/VillageResolver.java, village/VillageTracker.java, src/api/java/com/selluastar/fealty/api/*, docs/API.md, gametest/FealtyGameTests.java, and data/fealty/fealty/quest_chains/rare_villager.json.
- Today: the rare_villager chain starts when a Trusted player asks an elder about rumours. Each step is given by a different villager (CHAIN_ROLE attachment), then comes a map to a hidden hamlet, a Keeper trial and a final reward.

BUILD THESE EIGHT FEATURES

1. Single-giver chains. Add a field "giver" to QuestChainDefinition ("per_step" by default, or "single"). With "single", one villager holds one role (field "giver_role", optional "giver_title" component) and offers every step in order through ChainManager, using the existing step and quest fields.

A step may carry an optional "unlock" condition (see feature 4).

rare_villager.json must behave exactly as before.

2. Stable extension API. Move or mirror into the API jar (package com.selluastar.fealty.api.quest) the types other mods need: QuestType, QuestObjective, QuestContext, QuestGiver and the quest-type registry key. Add a new registry "fealty:chain_kind" with a ChainHandler interface (start, onStepComplete, onChainComplete) so other mods can define chain kinds beyond RARE_VILLAGER and GUILD. Document it in docs/API.md.

3. Rumour hook. New data folder data/<ns>/fealty/rumours/<id>.json:

```
{ "chain": "ns:chain_id", "stage": 1, "min_tier": "fealty:trusted",
  "weight": 5, "professions": [], "lines": ["ns.rumour.key"], "requires": {
  condition }, "grants": [ effect ] }
```

Add a RumourService. It rolls once per villager per in-game day on each conversation (hook into the node build in VillagerDialogue). Config: rumour_chance (default 0.03), pity_step (0.01), pity_cap (0.12), guarantee_after (30 eligible conversations). A successful roll picks a weighted eligible rumour, adds its line to the villager's dialogue, applies "grants" and records progress per player. The elder's existing "rumours" action should draw from the same pool. Fire a RumourEvent (cancelable, lines editable).

4. Branching conversations from data.

data/<ns>/fealty/conversations/<id>.json (format below), compiled into the existing DialogueNode and Option records and routed through DialogueService.choose. Add two registries, "fealty:dialogue_condition" and "fealty:dialogue_effect", with built-ins. Conditions: has_item, tier_at_least, flag, advancement, dimension, quest_stage, time_of_day, and all/any/not. Effects: take_item, give_item, set_flag, clear_flag, add_rep, start_quest, grant_advancement, run_function. A reply whose condition fails is shown disabled with its "hint" text. DialogueScreen must support up to 6 replies.

5. Player flags. Add a PlayerFlags attachment (ModAttachments) and RepApi methods getFlag, setFlag and clearFlag (ResourceLocation keys). Persist across death and relog. Add /fealty flag <player> get|set|clear for debugging.

6. Dialogue contributors. Add a DialogueBuildEvent fired in VillagerDialogue.node and GiverDialogue.node with getPlayer, getNPC, addOption and addLine. Options whose id starts with "<modid>:" are routed back to the listener that added them when the player picks them.

7. Special-role villagers and villages. API methods: assignRole(villager, chainId, role, title) built on the existing CHAIN_ROLE data, findVillager(server, village, predicate), and findVillage(server, origin, minDistance, maxDistance, predicate) built on VillageResolver and VillageTracker. The assigned villager shows its title in its name display.

8. Events in `api.event`: `RumourEvent`, `DialogueEvent` (Open, Choose) and `ChainStageEvent` (Start, Advance, Complete), as plain NeoForge events that anything can subscribe to. Do not add or change any integration with another mod.

CONVERSATION FORMAT (feature 4)

```
{
  "id": "ns:npc/first_meeting",
  "speaker": "ns:role",
  "start": "greet",
  "nodes": {
    "greet": {
      "text": "ns.dialogue.greet",
      "replies": [
        { "label": "ns.reply.show_letter", "if": { "has_item": "ns:letter" } },
        { "label": "ns.reply.bye", "end": true }
      ]
    },
    "letter": {
      "text": "ns.dialogue.letter",
      "effects": [ { "take_item": "ns:letter" }, { "set_flag": "ns:met" } ],
      "replies": [ { "label": "ns.reply.ready", "goto": "greet" } ]
    }
  }
}
```

RULES

- Backwards compatible: existing data, saves and the `rare_villager` chain behave exactly as before. Bump `api_version` to 1.3.0 (`gradle.properties` and `FealtyApi.API_VERSION`).
- Match the code style: records with codecs, `DeferredRegister` registries, NeoForge events, no dependency of any kind on another mod. Leave the existing compat modules exactly as they are.
- Fealty must start and play fully with no other mod installed. Nothing you add may name another mod, including in code, data, lang keys, docs, examples and tests. Use the neutral namespace "example" in all examples and tests.
- The server decides and the client displays. Anything the client needs goes through a payload like `OpenDialoguePayload`.
- Add lang keys to `assets/fealty/lang/en_us.json` for any text Fealty itself shows.

DELIVERABLES AND ACCEPTANCE

- Update docs/API.md with every new type, registry, event and data format.
- Add an example data pack in docs/examples/single_giver_chain/ with a rumour, a conversation and a single-giver chain.
- Add GameTests in FealtyGameTests: rumour roll and pity, a single-giver chain end to end (rumour, letter, three steps, reward), a contributor reply, and flags surviving relog.
- Run the build and the tests, then list what changed and anything you chose not to do.
- If something is ambiguous, pick the simplest option that keeps the API general, and note it.

17. Advancements and FTB Quests

Build tags. [CC] every advancement, custom stat, tag and skyseam: id below, each fired by a custom or vanilla trigger from the system it belongs to, with a GameTest that fires it. [FIRST PASS] advancement titles, descriptions and icons (icons reuse item sprites). [MANUAL] the FTB Quests chapters, quests and rewards: the author builds them from these ids, and no starter chapter is generated (decision 9 in section 23).

Every milestone is an advancement under the skyseam: namespace, plus custom stats and tags, so the author's questbook needs no custom code. The FTB Quests source was checked: its task types are item, custom, xp, dimension, stat, kill, location, checkmark, advancement, observation, biome, structure, gamestage, fluid and interaction. The table names the best fit for each milestone. Fealty already ships its own FTB Quests rep task, which can sit beside these.

Advancements

Ids are under skyseam: . A star means a group of several, such as one per biome.

Advancement	Fires when	Best FTB task
rumour/whisper	A villager mentions the scar	Advancement
rumour/lead	Survey Map received	Item
rumour/introduction	Starlit Letter received	Item
stargazer/met	First talk with Wylan	Advancement

Advancement	Fires when	Best FTB task
favor/survey, favor/three_seals, favor/silent_chapel, favor/ring_run, favor/stormglass	Each favor finished	Advancement
favor/all_five	All five favors done	Advancement
cairn/found, chapel/found	Enter the structure	Structure
storm/stormlings	10 Stormlings killed	Kill
aperture/crafted	Harmonic Aperture crafted	Item
aperture/mounted	Aperture placed on a ship	Advancement
seam/opened	First Seam opened	Stat seams_opened
seam/crossed	First crossing	Dimension
halcyon/anchorage	Reach the Sundered Obelisk	Structure
halcyon/biome_* (7)	Enter each biome	Biome
halcyon/tide_* (4)	Live through each Tide	Advancement
halcyon/rebound	Survive a Rebound	Advancement
halcyon/underbelly	Reach an Underbloom hollow from below	Advancement
halcyon/wreck_* (3)	Salvage each wreck size	Structure
halcyon/logbook_6, logbook_12	Pages found	Stat logbook_pages
helpers/moonsilk, whale_song, antler_glass	First of each helper product	Item
helpers/cogwren_page	First page from a Cogwren	Advancement
helpers/wrightling_friend	Friendly tier with Wrightlings	Fealty rep task
boss/regent, boss/skyray, boss/cantor	Each Guardian defeated	Kill

Advancement	Fires when	Best FTB task
<code>boss/skyray_boarded</code>	Stand on the Skyray's back	Advancement
<code>boss/skyray_airborne</code>	Beat it without touching ground	Advancement
<code>boss/cantor_silent</code>	Beat it without a Listener noticing you	Advancement
<code>boss/all_three</code>	All three Guardians	Advancement
<code>coupler/crafted</code>	Pneumatic Coupler crafted	Item
<code>coupler/first_lift</code>	Move a ship with it	Advancement
<code>sounder/crafted</code>	Echo Sounder crafted	Item
<code>sounder/first_ping</code>	First Pulse	Stat <code>sounder_pings</code>
<code>sounder/first_echo</code>	Reveal a Stitch	Advancement
<code>exit/found_all</code>	All four Stitches logged	Advancement
<code>exit/first_use</code>	First exit taken	Advancement
<code>exit/round_trip</code>	Three crossings each way	Stat <code>crossings</code>
<code>spire/floor_*</code> (4)	Each Spire floor solved	Advancement
<code>spire/stillpoint</code>	Glyph Cipher solved	Advancement
<code>challenge/pacifist</code>	Cross the Halcyon killing no hostile	Advancement
<code>challenge/hush_runner</code>	Cross the Hush without alerting a Listener	Advancement

Custom stats

`seams_opened`, `crossings`, `exits_used`, `moonsilk_harvested`, `whale_songs`,
`logbook_pages`, `sounder_pings`, `listeners_alerted`, `guardians_defeated`.

Tags

Item tags `#skyseam:favor_components`, `#skyseam:stay_relics`, `#skyseam:halcyon_materials`. Entity tags `#skyseam:guardians`, `#skyseam:hostile`, `#skyseam:helpers`. Structure tag `#skyseam:halcyon_landmarks`. FTB Quests item and kill tasks accept tags, so one quest can cover a whole group.

Suggested chapters

These are suggestions for the author's questbook (`[MANUAL]`). Claude Code does not create chapter files.

1. **Whispers:** rumour, rep with villages, meeting the Stargazer.
2. **Five Favors:** one quest per favor, then the Keystone.
3. **Unpicking the Sky:** craft, mount and open the Seam.
4. **Arrival:** biomes, Tides, helpers, Rebound.
5. **Stay Guardians:** three bosses in any order, plus the challenge variants.
6. **Craftsmanship:** the Coupler and the Sounder.
7. **Listen:** find the four Stitches and go home.
8. **The Spire:** four floors and the Cipher.
9. **Challenges:** optional, hidden until the main line is done.

18. Idea pool

Build tags. Backlog. None of this is required and none of it is part of the first build. Build an item only when the author names it. An item built later is `[CC]` for its logic and `[FIRST PASS]` for its art and sound, except where it is tagged inline.

More than 80 extra ideas, none required, grouped so the author can pull a few per update. Each is built from the core systems above.

Creatures

- **Prism Lamprey:** an eel that swims along wind lanes and dive-bombs ships at dusk. Drops Prism Dust.
- **Cloud Tortoise:** a huge, slow neutral animal with a garden on its back. Moor to it and Glass Berries make it carry you along a lane.

- **Gilded Kite:** an ally bird called with a whistle that scouts 200 blocks and marks creatures for 10 seconds (5 minute cooldown).
- **Hush Moth:** eats noise. Carried in a Quiet Jar it mutes footsteps in a 6-block radius.
- **Tide Heron:** wades the Mirror Sea, and its posture predicts the next Tide.
- **Echo Finch:** repeats the last bell note it heard, a hint source in the Belfry.
- **Dewfox:** a shy fox that leads you to Moth Groves if you leave Glass Berries.
- **Wrightling Elder:** a rare, seasonal quest-giver with a longer story.
- **Brass Beetle:** rolls Prism Dust into balls you can collect.
- **Stray Child:** a Stray Reflection variant that only copies, never attacks, used in a puzzle.

Items and tools

- **Petal Kite:** a one-handed glider that rides wind lanes.
- **Quiet Jar, Lantern Whistle, Tide Compass:** a noise muffler, a moth caller, a pointer toward the next Tide.
- **Prism Lens:** a Sounder upgrade with more range for more cells.
- **Cell Charger:** a block that recharges Resonance Cells from Create rotation speed.
- **Skyfarer's Journal:** records your flight path as a line on the Almanac Chart.
- **Glass Berry Jam:** a food that briefly boosts ping range.
- **Rope Ladder Kit:** quick deck access for large ships.

Blocks and building

- Cloud Glass in colors, Wright Brass trim set, Petal Thatch roofs, Lumen Lamps.
- Prism Windows that throw colored light patches on the floor.
- Chime Chains that play with the wind, Glassreed Thatch.
- Gravity Plates as functional decor for wreck-style rooms.

Puzzles

- **Tide Clock:** an orrery GUI where you set the Lantern-Sun position to open a vault.
- **Whisper Walls:** hidden doors that only answer to pings.
- **Coupler Docks:** drag a floating platform into a socket with the Pneumatic Coupler. A puzzle designed around the reward.
- **Moth Maze:** lead a Lanternmoth swarm with a lantern to a switch.
- **Shadow Sundial:** time the Lantern-Sun so a shadow lands on marks.

- **Windbell Chimes:** tune blade pitches so wind plays a melody that opens a door.
- **Weighted Scales:** balance pans using item weights.
- **Petal Lock:** plant flowers in a coded order.
- **Glyph Mirror:** glyphs read reversed in a mirror, so the cipher needs a second look.

Events and Tides

- **Aurora Night:** ribbons act as extra wind lanes and whales sing in chorus (double Whale Song).
- **Lantern Eclipse:** the orb dims for 3 minutes and Listeners walk in daylight. High risk and reward.
- **Glass Meteors:** Prismite showers leave temporary mining nodes.
- **Great Migration:** every whale crosses in one pod, and riding with them reveals all Stitches.
- **Hollow Tide:** all sound muted, pings are the only navigation. Hard mode.
- **Wreck Surge:** buried wrecks rise into reach for 10 minutes.
- **Pollen Storm:** low visibility, triple helper yields.
- **Seam Tremor:** a mini-seam flickers open with a loot cache beyond it.

Boss variants

- Tide-touched fights: one Tide's rules active during the fight.
- Cantor's Encore: longer phrases and visual-only clues.
- Spire boss rush with shared health, unlocked after the Stillpoint.
- Co-op scaling: +50 percent health and extra pylons per extra player.
- Time-trial stat and a no-damage advancement per boss.
- Phase-reactive layered boss music [SUPPORT ONLY]. The hook is built in section 21, and the tracks are the author's.

Structures and landmarks

- **The Orrery:** a working model of the pocket inside the Spire that doubles as the Tide Clock.
- **Choir Cathedral:** a ruined Hush hall with giant organ pipes that play when pinged.
- **Unfinished Flagship:** a half-built Wright ship at the Anchorage. Gather parts across the Halcyon to finish it and unlock a sky parade.

- **Lighthouse of Seams:** a tower whose sweeping beam highlights the nearest unfound Stitch.
- **Pearl Gardens:** glass biospheres of tropical plants in the Underbloom.
- **Whale Graveyard:** sacred bones and lore, with a sing site.
- **Floating Market:** a Wrightling barge that drifts between Moorings once a day.
- **Quiet Library:** a room of Wrightling books and a librarian.

Create and Aeronautics tie-ins

- Opal envelope paint crafted from Prism Dust, and Antler Glass propeller skins (cosmetic).
- Wrightling cargo contracts: deliver Create-made goods by ship to a Mooring for rep and rare materials.
- Wind lane turbines that power Create machines in a dimension with no rivers.
- Wright Skiff and other ships as Create schematics, found in logbook pages and printable with the Schematicannon.
- Sky salvage: tow wreck chunks home with ropes.
- Spire lift and Cell Charger both run on Create rotation.

Fealty and villagers

- 30 or more rumour lines, written per profession.
- After the chain, a Wright stall opens in the Stargazer's village and permanently sells Halcyon goods.
- Villages you helped hang lanterns on their roofs.
- The Stargazer's apprentice remembers your closing answer in later dialogue.
- Wrightling rep tiers appear in Fealty's own standings screen.
- Fealty's mailbox delivers letters from Wrightlings after you return.

Multiplayer and teams

- Progress scope switch: per player or per FTB team.
- Crew charge: each extra seated player shortens the Seam charge by 0.5 seconds, down to 2.
- Shared Mooring respawn for a team, PvP off in the Halcyon by default.
- Independent boss loot per player.

Accessibility and quality of life

- Colorblind-safe Prism Routing with shapes as well as colors.
- Visual ring and text subtitles for echo pings.
- Reduce-motion toggle for Seam shake and flash.
- Server option to skip a puzzle already solved once.
- Almanac hints in three levels: nudge, hint, answer.

Modpack extras

- KubeJS events for Seam open, cross and exit, Tide change and boss kill.
- Config presets: Casual, Standard, Hardcore.
- Data-driven Tide table, spawn tables and wreck loot, so pack makers add their own items.
- A generated FTB Quests chapter file for the quests folder, built from the advancement table. [CC] , but only if the author asks, because decision 9 says no starter chapter.

Polish

- Footstep chimes on Glassreed, per-biome ambient loops, a whale chorus.
- Seam colors shift per Tide (a renderer tint, not a custom shader).
- Photo mode hotkey: hides the HUD and freezes the Lantern-Sun.
- Arrival cinematic toggle: a slow pan as the Seam opens.
- Particle quality slider and per-phase lighting.

19. Technical notes and build order

Build tags. [CC] all of it. This section is the plan Claude Code follows, and where it differs from the Design Bible this version wins.

Everything here is buildable, but the ship crossing between dimensions is the one piece nobody has proven yet, so it is milestone zero.

Stack

- **Minecraft 1.21.1 on NeoForge with Java 21**, matching Fealty (NeoForge 21.1.209). Sable's add-ons list NeoForge 21.1.228 or newer, so confirm the highest requirement in the pack.

- **Required dependencies:** Create, Create Aeronautics (which brings Simulated, the mod that owns the physics staff), Sable (which lists Create and Veil as its own requirements), **Fealty** (API 1.3.0, built by the Fealty session in section 16) and GeckoLib 4.
- **Where to get them:** Sable publishes its API at `https://maven.ryanhcode.dev/releases` as `dev.ryanhcode.sable:sable-common-1.21.1`. GeckoLib is `software.bernie.geckolib:geckolib-neoforge-${minecraft_version}:${geckolib_version}` from `https://dl.cloudsmith.io/public/geckolib3/geckolib/maven/`. Use the 4.x line for 1.21.1 and confirm the exact version there, because a 5.x alpha also exists. For anything without a Maven of its own, use Modrinth's Maven (`maven.modrinth`) or CurseMaven, and take the coordinates from the project's own page rather than guessing. Fealty comes from the author's local build of the Fealty project.
- **Soft dependencies:** KubeJS and FTB Quests (no code link, advancements only).
- **Animated mobs:** GeckoLib 4 for NeoForge 1.21.1, with models and animations authored in Blockbench.
- **Licenses:** Sable is under the Polyform Shield license. The Simulated project's code is MIT but its art assets are all rights reserved. Link to public APIs only and never copy their models or textures. The author checks both before redistributing a modpack.

What is easy and what is risky

Feature	Effort	Risk	Note
Seam visuals (entity, renderer, particles)	Medium	Low	Stays shader-pack safe by avoiding a true portal render
Entry trigger (Aperture, ship detection, ground check)	Medium	Medium	Needs Sable calls to find the sub-level that holds the block and its pilot
Ship crossing between dimensions	High	High	Spike first. Routes A and B in section 6
Bounded floating-island dimension	High	Medium	Custom chunk generator, Veil deflection
Halcyon physics data pack	Low	Low	One JSON file

Feature	Effort	Risk	Note
Three animated bosses	High	Medium	Boarding the Skyray is the hardest part. Fallback: weak points hit from ships or the air
Echo Sounder visuals and noise events	Medium	Low	Ring and beam rendering plus a server noise bus
Six custom screens	Medium	Low	Match Fealty's look
Eight Fealty features ([OTHER REPO])	Medium	Low	Separate task, prompt in section 16
Advancements, stats, tags	Low	Low	Data generation
Pneumatic Coupler	Low	Low	Recipe only
Models, textures, structures, sounds ([FIRST PASS])	High	Medium	Buildable by script and MCP, but only the author can judge the result

Build order

Each milestone ends with something the author can launch and film, and with `docs/STATUS.md` updated. A milestone is done when `./gradlew build`, `runData` and `runGameTestServer` pass and a headless `runServer` boots. Every milestone also includes the first-pass art and sounds for its own features, in the order given in section 20.

ID	Milestone	What is built	Done when
M-1	Environment and skeleton	<code>docs/ENVIRONMENT.md</code> , the project with every dependency, the four run configs, <code>SableBridge</code> and <code>ExternalIds</code> , the Blockbench MCP servers registered, the <code>docs/</code> files created	The build passes and an empty mod boots on a headless server
M0	Spike	Move a test Sable ship between two dimensions with Route A, then Route B. If neither works, adopt the fallback in section 23	The result and chosen route are in <code>docs/DECISIONS.md</code>

ID	Milestone	What is built	Done when
M1	The Seam	Project content skeleton, the <code>/skyseam</code> debug commands, the Seam entity, renderer and particles, the full reveal, synthesized reveal sounds. Look and sound are [FIRST PASS]	The command opens and mends a Seam, and the author can film it
M2	Entry	Harmonic Aperture block and block entity, trigger rules, gauge HUD, Skychart, ship transfer	A GameTest per trigger rule, and a test ship crosses
M3	The shell	Dimension, sky, physics pack, Arrival Lane, Veil, Mirror Sea Rebound, biomes, the structure generator tools and the Sundered Obelisk	A ship arrives and the Obelisk is in view
M4	Systems	Tides, wind lanes, Moorings, Almanac skeleton, the noise bus, the music manager with every slot empty ([SUPPORT ONLY]), the audio synthesis tool	Tides cycle, ships dock, the game runs silent of music without errors
M5	Helpers	The four helper species and the Wrightling as placeholder-then-first-pass creatures, materials, the Pneumatic Coupler recipe	The Coupler can be crafted early from creative-spawned parts
M6	Regent	Prism Regent, Refraction, Hall of Reflections, Standing Arch	The fight is winnable and drops its relic
M7	Skyray	Tempest Skyray, Gale Ring, boarding, Pneumatic Heart	The fight is winnable and completes the Coupler
M8	Cantor and exits	Hollow Cantor, Echo, the Echo Sounder, Stitches, soft-lock nets	The first full loop: enter, fight, leave
M9	The Spire	Four floors, puzzles 5 to 12 as needed, the Glyph Cipher	The Stillpoint Heart can be earned

ID	Milestone	What is built	Done when
M10	The chain	Needs Fealty 1.3.0. Rumour data, the Stargazer, the five favors and their puzzles, the closing talk, Wrightling dialogue and services	Part 1 plays start to finish
M11	Questbook	Advancements, stats and tags (ids only, no chapter file, decision 9)	A GameTest fires every advancement
M12	Polish and handoff	Remaining sounds, structure and texture review pass, accessibility options, config presets, <code>TODO-MANUAL.md</code> finalized from section 22	The author has a complete list of what is left for them

The Fealty session (section 16) should start on day one in parallel. It only blocks M10.

Risks and answers

- **API drift:** wrap every Sable and Aeronautics call in one `SableBridge` class and pin versions, so an update is a one-file fix.
- **Intrusive mixins:** test against the author's real pack early, not at the end.
- **Big-ship hitch on crossing:** cap size and let the pearl-white flash hide the load.
- **Multiplayer sync:** the server decides everything and the client only plays visuals from packets.
- **Server stops with a Seam open:** store Seam state in world data and mend on load.
- **Blockbench MCP is experimental:** if it fails, generate the geo and animation JSON by script and record that in `docs/ENVIRONMENT.md`.
- **Testing:** Claude Code can write code, data, models, textures, structures and synthesized sounds, and can run the build, data generation and headless GameTests. It cannot launch Minecraft and look at it or listen to it, so each milestone ends with the author running it and sending logs or screenshots. `docs/STATUS.md` says what to look at.

20. Asset manifest

Build tags. `[FIRST PASS]` for every model, texture, animation, sprite and particle below, and for the sounds as marked in the sound table. `[CC]` for spawn eggs, `.mcmeta` files, `sounds.json`, lang entries and the renderer-only pieces that have no art. `[SUPPORT ONLY]` for music. The author's final polish pass on the hero pieces is `[MANUAL]` and is listed in section 22.

Everything that has to be modeled, drawn, animated, built or recorded, with who makes it. Rig and animation names are working names under the `skyseam` namespace. Sizes marked "suggested" are the designer's guesses; everything else comes from the sections above.

At a glance: 18 creature rigs (3 bosses, 15 mobs), 3 small entity models, 22 block entries, 33 items (32 need new art), 9 screens, 14 particles and about 150 sounds. Structures are generated by script (section 14) and are not Blockbench work.

Where assets live (the swap rule)

Claude Code writes this table into `docs/ASSET-PATHS.md`. Replacing any file with a file of the same name must work with no code change.

Kind	Folder under <code>src/main/resources/</code>	Naming
Entity geometry	<code>assets/skyseam/geo/</code>	<code><name>.geo.json</code>
Animations	<code>assets/skyseam/animations/</code>	<code><name>.animation.json</code>
Entity textures	<code>assets/skyseam/textures/entity/</code>	<code><name>.png</code> and <code><name>_glowmask.png</code>
Block, item and GUI textures	<code>assets/skyseam/textures/block/</code> , <code>item/</code> , <code>gui/</code>	Moving ones add <code><name>.png.mcmeta</code>
Particles	<code>assets/skyseam/particles/</code> and <code>textures/particle/</code>	<code><id>.json</code>
Sounds	<code>assets/skyseam/sounds/<group>/</code>	<code><name>.ogg</code> , declared in <code>assets/skyseam/sounds.json</code>
Music	<code>assets/skyseam/sounds/music/</code>	<code><slot>.ogg</code> (section 21)

Kind	Folder under <code>src/main/resources/</code>	Naming
Structures	<code>data/skyseam/structure/</code>	<code><name>.nbt</code>
Lang	<code>assets/skyseam/lang/</code>	<code>en_us.json</code>

Source models stay in `art/models/<name>.bbmodel` outside the resources folder, and generator scripts stay in `tools/`.

Production rules

- **Mobs and bosses** are built with the Blockbench MCP (section 2) in GeckoLib mode, kept as `.bbmodel` sources, and exported to `.geo.json` plus `.animation.json`. If no MCP works, generate the same files by script. Suggested texture sizes: 64 by 64 for small mobs, 128 by 128 for mid-size mobs, the Regent and the Cantor, 256 by 256 for the Whale and the Skyray. Box UV is the simplest layout for generated textures.
- **Glow** comes from a separate `_glowmask` texture per model (GeckoLib's `AutoGlowingGeoLayer` convention), so lamps, crowns and bells shine in the dark. Make it by script: copy the base texture, then clear every pixel that should not glow.
- **Shared animation names** on every mob that has the pose: `idle`, `walk`, `fly`, `attack`, `hurt`, `death`. The tables below list only the extra ones.
- **Blocks and items** use 16 by 16 textures in the pastel-glass palette of screenshot 2, generated by scripts in `tools/textures/` that share one palette file so the whole set stays coherent. Moving textures (Mirror Sea, Prismite, Lumen Lily) use `.mcmeta` frame files.
- **Animated blocks** (Aperture, Star Dial, Glyph Cipher) are block entities with GeckoLib renderers. The Seam has a custom renderer and no model.
- **Sounds** are mono `.ogg` files listed in `sounds.json` with subtitles, because only mono sounds are positional.
- **Placeholders first.** Every entity gets a single-box model and a flat color before any art, so gameplay and quests can be built and tested while the art catches up.
- **Look at the result.** Open every generated texture and every model render as an image before reporting it.

Boss rigs

All three are `[FIRST PASS]`.

Rig	Size	Main bones	Animations	Textures and notes
Prism Regent	about 4 blocks (suggested)	root , torso , head , crown (five spikes), arm_l and arm_r (upper and lower), core , ribbon_1 to ribbon_8 (three links each)	float , split , decoy_pop , overexposure_charge , overexposure_release , stunned , shatter_phase , death	regent plus glowmask, and a translucent regent_decoy skin on the same rig. The floor reflection is a renderer effect drawn for the real Regent only
Tempest Skyray	48 blocks across	root , plate_1 to plate_8 , wing_l1 to wing_l4 , wing_r1 to wing_r4 , head , tail_1 to tail_6 , sac_1 to sac_6 , heart , emitter_l , emitter_r	glide , bank_l , bank_r , throw_cell , cyclone_surge , low_pass , sac_inflate , sac_pop (once per sac bone), heart_exposed , stagger , death_breakup	skyray plus glowmask. Needs multipart hit boxes (head, six sacs, heart, tail) like the Ender Dragon, and solid back plates so players can stand on it during the low pass

Rig	Size	Main bones	Animations	Textures and notes
Hollow Cantor	about 3.5 blocks (suggested)	root, robe_1 to robe_3, torso, head with hood, arm_1 and arm_r, baton, bell_ring pivot with bell_1 to bell_12	sway, shimmer (the near-invisible state), ping_reveal, conduct_4, conduct_6, stunned, summon_choir, crescendo, death_toll	cantor plus glowmask for the bells, and a cantor_outline skin shown only while pinged

Mob rigs

All fifteen are [FIRST PASS] .

Mob	Main bones	Extra animations	Notes
Lanternmoth (palm-sized)	body, head, antenna_1 and antenna_r, wing_fore_1 and _r, wing_hind_1 and _r, lamp	hover, flap, rest	Glowmask on lamp, dimmer in the Hush. The cocoon is a block, listed below
Wayfarer Whale (14 blocks)	head, jaw, body_1 to body_3, tail_1, tail_2, fluke, fin_1 and fin_r, ring_1 to ring_3, throat	glide, sing, turn, roost	256 texture. Glowmask pulses during sing. Hit box smaller than the model
Cogwren (chicken-sized)	body, head, beak, wing_1 and wing_r, tail, leg_1 and leg_r, gear	peck, flap_glide, carry, nest	gear spins faster while a Logbook Page is carried

Mob	Main bones	Extra animations	Notes
Bloomstag	body , neck , head , antler_l and antler_r (three tines each), four legs, tail	graze , shed , regrow , bloom_glow	Antler bones scale to zero during shed . Glowmask on the antlers
Wrightling (knee-high)	head with goggles, body , apron , arm_l and arm_r , leg_l and leg_r , pack , tool	work , trade_nod , wave , sit	Apron color variants for stall, dock and camp crews
Glasskipper	body , tail , fin_l and fin_r , jaw	skim , scatter	Translucent texture. Drawn in schools, so keep the cube count low
Updraft Gull	body , head , beak , wing_l1 , wing_l2 , wing_r1 , wing_r2 , tail , foot_l and foot_r	glide , flap , circle , perch	Circling is driven by code around wind lanes, so circle only needs a banked glide
Brasshorn Ram	body , head , horn_l and horn_r , four legs , tail	graze , headbutt_windup , headbutt	Brass horns get a painted gloss, no glowmask
Shardling	body , head , shard_crown , arm_l and arm_r , hand_shard , leg_l and leg_r	hop , throw_shard , shatter	Death plays shatter and bursts Prism Dust particles
Hollow Listener (tall)	hips , torso , leg_l and leg_r (long), arm_l and arm_r , head , ear_l and ear_r	stalk , sneak , perk_ears , charge , swipe	No eyes. perk_ears swivels the bell ears toward a noise. Glowmask on the ear rims

Mob	Main bones	Extra animations	Notes
Thread Spinner	abdomen , thorax , head , leg_l1 to leg_l4 , leg_r1 to leg_r4 , spinneret	climb , cast_thread , bite , curl	The golden thread is a line the renderer draws between two anchors, not a model. Glowmask on the abdomen
Gale Harrier	body , head , beak , wing_l1 to wing_l3 , wing_r1 to wing_r3 , tail , talon_l and talon_r	glide , flap , dive , strike , retreat	Feather bones on the wing tips flutter in the wind
Stray Reflection	head , body , arm_l and arm_r , leg_l and leg_r , shard_1 to shard_6 floating around it	walk_water , strike , shatter	Translucent mirror texture. The copy of your movement a second late is code, so the model only needs biped poses
Pollen Bloat	bell , tentacle_1 to tentacle_6 (three links each), sac_1 to sac_3 , core	drift , inflate , burst	Glowmask on the core. Pollen particles on burst
Stormling (small)	core , arc_1 to arc_4 , crown	float , charge , arc_strike , grounded , fizzle	Glowmask. Arcs are renderer lines between the Stormling and its target. grounded plays when a lightning rod catches the arc

Other entity models and renderers

Piece	What it is	Notes
Wind Ring	An eight-segment ring, about 6 blocks across (suggested)	States <code>dormant</code> , <code>next</code> , <code>passed</code> , <code>missed</code> . Pulses while it is the next ring. The Ring Run uses 12. <code>[FIRST PASS]</code>
Charged Cloud Cell	Skyray projectile	Small sphere with arcing lines and a <code>spin</code> loop. <code>[FIRST PASS]</code>
Glass Shard	Shardling projectile	Tiny spinning shard in the Mirror Shard material. <code>[FIRST PASS]</code>
The Seam	Custom renderer, no Blockbench model	Additive line-strip edges, translucent quads, the pre-baked interior sky with parallax, golden thread strips and scar particles (see section 6). <code>[CC]</code> code, <code>[FIRST PASS]</code> textures
Prism beams and sonar rings	Renderer effects	Additive quads and ring meshes, no model. <code>[CC]</code>

Blocks

All block art is `[FIRST PASS]` ; the block entities and behavior are `[CC]` .

Block	Build	States and animation
Harmonic Aperture	GeckoLib block entity	<code>idle</code> , <code>spin_up</code> , <code>charged</code> , <code>cooldown</code> . The core ring and three status gems glow
Prism Pylon	Block entity renderer	Head turns in 15 degree steps. Beam socket lights when hit
Prism Splitter	Static model plus beam renderer	Crystal with three output arms, lit when fed
Prism Receiver	Static	Red, green and blue lenses, lit and unlit

Block	Build	States and animation
Tide Vane	Block entity renderer	One pose per Tide: Calm, Prismfall, Gale, Bloom, Stillness
Mooring Post	Static plus rope	Docked and free rope states
Wren House	Static	Empty and occupied
Star Dial	GeckoLib block entity	Three rings turn independently, plus a solved pose
Bell Frame	Static with a clapper slot	Empty, clapper placed, ringing (three in the Silent Chapel)
Bell Pillar	Static, large	Ringing animation per pitch (the ring of them in the Belfry)
Echo Bell	Static	Hidden outline and revealed (three in the Hush caves)
Thread Loom	Static plus GUI	Idle and threaded
Glyph Cipher	GeckoLib block entity	Wheel turn, three relic sockets (empty or filled), open pose
Float Plate and Sink Plate	Static	Pressed and unpressed. Glow color tells them apart
Mirror wall	Static, translucent	Reflective wall tile for the Mirror Maze
Spire Lift car	Moving platform	idle , rise , locked
Moth Cocoon	Static	Four growth stages, from empty to ready to shear
Glass Berry bush	Crop-style	Three growth stages
Stillstone set	Static set	Stone, bricks, slabs, stairs, walls, pillars, glass
Island flora	Cross and plant models	Petal Carpet, Cloudmoss, Glassreed (two high), Lumen Lily, Prismite clusters in three sizes, lantern vines, glow moss, hanging roots
Cloud blocks	Static	Soft, slightly translucent, in a few shades

Block	Build	States and animation
Mirror Sea	Animated surface	Reflective and nearly flat, scrolling .mcmeta frames

Items

Group	Items	Art
Tools (8)	Pneumatic Coupler, Echo Sounder, Skychart, Almanac, Tuning Fork, Prism Spyglass, Halcyon Lantern, Resonance Cell	16 by 16 icons. The Echo Sounder and the Halcyon Lantern also get held 3D models, and the Sounder gets pulse and beam use poses. The Coupler needs no art under Option A (the original model, renamed)
Quest papers (2)	Survey Notes, Flight Logbook	Icons
Materials (13)	Moonsilk, Prism Dust, Mirror Shard, Gale Plume, Antler Glass, Golden Thread, Hollow Shell, Whale Song Shell, Pneumatic Heart, Resonant Tine, Muted Bell, Logbook Page, Glass Berries	Icons. The Pneumatic Heart and the Whale Song Shell get a faint idle glint
Favor parts (6)	Star-Glass Lens, Triple-Wax Seal, Tuned Clapper, Windwheel Gear, Stormglass Vial, Skywright's Keystone	Icons. The Keystone is the hero asset, with a slow glint animation
Relics (4)	Prism Core, Gale Core, Echo Core, Stillpoint Heart	Icons with an idle glow. The three cores are also drawn inside the Glyph Cipher sockets
Spawn eggs (18)	One per mob	Vanilla template with two tint colors per mob, no new art. [CC]

All icons are [FIRST PASS] .

Screens

Screen	Textures needed
Dialogue	None new, since Fealty draws the box. Portraits for the Stargazer and the Wrightling if the box does not draw the entity
Aperture	Panel, status ticks, charge ring, Skychart slot
Seam Gauge HUD	Compact bar with three ticks and a distance readout
Almanac	Page background, five tab icons, entry frame, map overlay for the Chart tab
Star Dial	Star field background and three ring layers
Thread Loom	Grid, node sprites and thread sprites
Glyph Cipher	Wheel, three sockets and the glyph set
Echo Sounder radar	Plan-view disc, sweep ring, blips and compass ticks
Wrightling services	Panel plus four icons: Trade, Lost and Found, Tide forecast, hints

All screen art is [FIRST PASS]. Glyphs are drawn once and reused as the carved wall tile, the GUI sprite and the Almanac entry. Advancements need one tab background, and their icons reuse item sprites.

Sky, world and particles

- **Lantern-Sun:** a disc with rays, plus the phase art the Flower Clock reads. **Sky:** day colors per Tide as dimension-effects data, a star layer, aurora ribbons for the Hush, and one cloud texture. Fog is data per biome. [FIRST PASS]
- **Particles (14):** seam_mote, seam_spark, thread_snap, scar, prism_dust, shard_glint, petal, wind_mote, pollen, sonar_ring, ping_ripple, lantern_glow, bell_wave, arc_spark. [CC] types and providers, [FIRST PASS] textures.

Sounds

Every sound below exists in `sounds.json` from the milestone that needs it, with a subtitle key, pointing at a stand-in until a better file replaces it. Section 21 explains the methods.

Group	Sounds	About	Method and tag
The Seam	Hairline tink, tearing crack, thread snap in five pitches, open hum loop, ring pulse, crossing whoosh, closing chime	11	Synthesized mono OGG. <code>[FIRST PASS]</code> , and the first candidates for the author's replacement
Ambience	One loop each for Calm, Prismfall, Gale, Bloom, Stillness and the Hush	6	Synthesized seamless loops (filtered noise and slow tones). <code>[FIRST PASS]</code>
Music	Five layers of sparse piano and glass harmonica, one per Tide, plus one theme per boss	8	<code>[SUPPORT ONLY]</code> . Slots only, no files
World	Glassreed chime in five pitches, Lumen Lily bounce, petal rustle, Prismite chime, Mirror Sea splash and rebound	10	Vanilla stand-ins with pitch changes, chimes synthesized. <code>[FIRST PASS]</code>
Creatures	Ambient, hurt and death for each of the 15 mobs, plus specials: Whale song in four variants, Cogwren gear tick, antler shed chime, Listener ear ring, Spinner thread twang, Pollen burst, Stormling arc	55	Vanilla stand-ins per mob with pitch and volume changes. Specials synthesized. <code>[FIRST PASS]</code>
Bosses	Regent (split chime, overexposure whine, stun shatter, phase break, death), Skyray (cry, cell throw, cyclone loop, three pitched sac pops, heart exposed, death), Cantor (six bell notes, phrase fail buzz, summon, crescendo loop, death toll)	23	Synthesized bells and stingers, vanilla stand-ins for cries. <code>[FIRST PASS]</code>

Group	Sounds	About	Method and tag
Tools	Echo Sounder (pulse, beam loop, three return echoes, empty click), Aperture (charge loop, ready chime, mount click), Tuning Fork, Muted Bell, Almanac page turn, Skychart unfold, Mooring dock and undock	15	Synthesized plus vanilla stand-ins. [FIRST PASS]
Puzzles	Star Dial click, Bell Sequence strikes in four pitches, Ring Run pass and miss, Pylon turn and beam hum, Flower Clock bloom, Loom connect and fail, Gravity plates, Glyph Cipher socket and open, Lift start and stop, Turbine spin	19	Vanilla stand-ins (bell, piston, note-style tones) with pitch, plus synthesized chimes. [FIRST PASS]

Suggested art order

1. Placeholder boxes for everything, so quests and fights can be tested early.
2. The Seam: renderer textures, interior sky and the reveal sounds. It is the first moment players see.
3. The three hero props: Aperture, Echo Sounder and Almanac.
4. The Anchorage set: Stillstone, Mooring Post, Wrightling, Lanternmoth and Whale.
5. The bosses in the order players meet them: Regent, Skyray, Cantor.
6. Hostile mobs, flora and the remaining icons, then the sound pass.

21. Audio: built now, added later

Build tags. [FIRST PASS] sound effects. [SUPPORT ONLY] music. [MANUAL] the final sound design for the hero moments (section 22).

The straight answers

- **Sound effects: yes, as a first pass.** Claude Code registers every sound, writes every subtitle and `sounds.json` entry, points them at vanilla stand-ins, and synthesizes the hero sounds by script. It cannot hear any of it, so the author judges the result.

- **Music: no.** Claude Code cannot compose music. It builds the slots and the music manager so tracks can be added later with no code change, and the game runs silent of music until then.
- **An audio-generation MCP or API: optional.** Nothing here depends on one.

How sound effects are covered

1. **Registry and `sounds.json`**. Every sound event is registered in code, has an entry in `assets/skyseam/sounds.json` and a subtitle key in the lang file. All 147 sounds in section 20 exist from the milestone that needs them.
2. **Vanilla stand-ins.** An entry can point at a vanilla sound event with `"type": "event"` plus its own `volume` and `pitch`. That gives every creature, block and tool a sound on day one. Replacing a stand-in later means changing that one entry to name a file. No code changes.
3. **Synthesized files.** `tools/audio/synth.py` (Python with numpy) builds the hero sounds from simple recipes: plucked strings for thread snaps, inharmonic bell tones for chimes and bell notes, filtered noise sweeps for cracks and whooshes, layered sine hums for loops. It writes mono Ogg Vorbis files (ffmpeg with libvorbis, or the soundfile package) into `assets/skyseam/sounds/<group>/`. Loops are crossfaded so they join cleanly.

Checks Claude Code can run

A script (`tools/audio/check.py`) verifies, for every synthesized file: mono, a common sample rate, a sane length, peak level below full scale, no DC offset, and for loops a small jump at the join. It also writes a spectrogram PNG that Claude Code can view. A second script verifies that every registered event has a `sounds.json` entry and a subtitle, and that every file an entry names exists.

These checks catch broken files, not bad taste. Creature voices and the whale song are the sounds a synthesizer will do worst, so expect them to be replaced. Everything labeled `[FIRST PASS]` goes in `docs/STATUS.md` as "needs your ears".

Music support `[SUPPORT ONLY]`

The intended sound is sparse piano and glass harmonica. The slots are:

Slot id	Used when	File name
<code>tide_calm</code> , <code>tide_prismfall</code> , <code>tide_gale</code> , <code>tide_bloom</code> , <code>tide_stillness</code>	The matching Tide is running (five layers, one per Tide)	<code><slot>.ogg</code>
<code>boss_regent</code> , <code>boss_skyray</code> , <code>boss_cantor</code>	That fight is active	<code><slot>.ogg</code>
<code>boss_<boss>_p1</code> , <code>_p2</code> , <code>_p3</code> (optional)	Per-phase stems for a boss, if present	<code><slot>.ogg</code>
<code>anchorage</code> , <code>spire</code> , <code>seam_open</code> , <code>hush_stinger</code> (optional)	The Anchorage, the Spire, the Seam reveal, a rare quiet stinger in the Hush	<code><slot>.ogg</code>

What Claude Code builds:

1. **Registered events.** One sound event per slot (for example `skyseam:music.tide_calm`), registered in code so biome and dimension data can name them.
2. **No music files.** None are shipped and none are faked. The manager checks whether a slot has a sound definition and plays nothing if it does not. No crash, and no repeated log spam (at most one info line per slot per session).
3. **A client music manager.** It chooses the slot from the context: an active boss fight first, then the current Tide, otherwise nothing. It hooks NeoForge's music selection event (`SelectMusicEvent`; confirm the name in the pinned version), crossfades between slots over about 3 seconds, follows the player's music volume, and has a `music.enabled` config switch. The choice itself is a pure function (context in, slot out) with unit tests.
4. **The Hush stays quiet.** In the Hush the manager plays no Tide music, only the optional `hush_stinger`, at most once every ten minutes.
5. **Boss stems.** The manager exposes the boss and its phase. If the phase stem exists it fades in for that phase. If not, the main theme plays through.
6. **Streaming.** Every music entry uses `"stream": true`.
7. **The how-to.** `docs/ADDING-MUSIC.md` explains: put the Ogg Vorbis file (stereo is fine for music) in `assets/skyseam/sounds/music/`, then either run `python tools/sync_music.py`, which scans that folder and regenerates the music block of

`sounds.json` , or paste the ready-made block from the document. A resource pack with its own `sounds.json` also works, so pack makers can add music without touching the jar. `TODO-MANUAL.md` lists each slot as `[SUPPORT ONLY]` .

Optional audio generation

If the author later connects an audio-generation MCP or API, Claude Code may use it only when asked, and only for stand-in sound effects. It never produces music unless the author supplies the files. The author checks the license of anything generated, and Claude Code notes the source of each file in `docs/AUDIO-SOURCES.md` .

22. Manual to-do list ("I'll add this later manually")

Build tags. `[MANUAL]` and `[SUPPORT ONLY]` items only. Claude Code seeds `TODO-MANUAL.md` from this section and keeps it current. Every item has a working stub, so none of them blocks a build or a test.

Guaranteed: "I'll add this later manually" `[MANUAL]`

These will be in the final mod, and Claude Code cannot make them.

#	Item	Why Claude Code cannot	What Claude Code leaves	Where the author's file goes
1	FTB Quests chapters, quests and rewards	The author designs the progression	Every id from section 17	The pack's FTB Quests folder, made in the FTB editor
2	Final sound design for the hero moments: the Seam reveal (11 sounds), the Echo Sounder (pulse, beam, echoes), whale song (4), the Cantor's six bell notes, boss cries, the Aperture charge	It cannot hear or record	Synthesized files or vanilla stand-ins under the final names	Overwrite the file in <code>assets/skyseam/sounds/<group>/</code> . If the entry names a vanilla event, change the entry to name the file

#	Item	Why Claude Code cannot	What Claude Code leaves	Where the author's file goes
3	Final art for the hero pieces: Harmonic Aperture, Skywright's Keystone, the Seam interior sky, the Almanac, the Echo Sounder held model, the three boss textures	It cannot judge beauty	First-pass models and textures with sources in <code>art/models/</code>	Edit the <code>.bbmodel</code> , export over the same names, or overwrite the texture
4	Polish on the hero structures (optional): Conservatory Spire, Sundered Obelisk, Standing Arch	It cannot judge quality	Generated <code>.nbt</code> files and <code>docs/STRUCTURE-MARKERS.md</code>	Build in creative, save with a structure block, overwrite the <code>.nbt</code> and keep the markers
5	Playtest tuning: difficulty, Tide timers, spawn rates, boss health, journey length	It cannot play the game	This document's defaults in config and data	The config file or data pack JSON
6	Real-hardware checks: shader packs, performance, multiplayer, dedicated server	It has no real session or GPU	A headless server boot and the GameTests it can write	The author sends logs and screenshots back
7	Mod identity: icon, page art, screenshots, trailer, changelog	Art and marketing	A plain placeholder <code>logo.png</code>	<code>src/main/resources/</code> and the mod page
8	License check and publishing	Legal and accounts	The license notes in section 23	The author

Optional later: support is built, content added if needed [SUPPORT ONLY]

Item	Hooks Claude Code builds	Added later by
Tide music (5 slots) and boss themes (3 slots)	Registered events, the music manager, the how-to (section 21)	Dropping .ogg files in
Optional music slots: anchorage , spire , seam_open , hush_stinger	The same manager	Dropping .ogg files in
Per-phase boss stems	The boss and phase exposed to the manager	Dropping .ogg files in
Other languages	All text through lang keys in en_us.json	Adding <lang>.json files

No voice acting is planned.

What TODO-MANUAL.md looks like

```
[MANUAL] Final Seam reveal sounds | assets/skyseam/sounds/seam/*.ogg | now: synthesized | owner: author
[MANUAL] FTB Quests chapters | pack config | now: ids only (section 17) | owner: author
[SUPPORT ONLY] Music slot tide_calm | assets/skyseam/sounds/music/tide_calm.ogg | now: silent | see docs/ADDING-MUSIC.md
```

23. Locked decisions and assumptions

Build tags. Reference. Claude Code follows these rows without asking. The author changes a row here before handing the document over if they disagree.

Nothing here blocks the build. Every question from the Design Bible has a locked answer that the rest of this document already uses.

Locked decisions

#	Decision	Locked answer	The alternative
1	What happens when you die in the Halcyon	Respawn at a Mooring Post. Items stay at the death spot, and Lost and Found replaces key items. A config switch turns the safety nets off	Respawn in the Overworld, which makes the trip a bigger risk
2	What if a ship cannot be carried between dimensions	Used only if the M0 spike shows neither route works. Players cross alone, and a Wrightling assembles a basic Wright Skiff for a fee, once per Tide cycle	Copy the ship block by block and accept that some contraptions will not survive (Route B)
3	Progress per player or per FTB team	Per player, because Fealty reputation is per player	Per team, which the Idea pool lists as a switch
4	Fealty	A required dependency. Skyseam compiles against Fealty API 1.3.0 and has no fallback dialogue runner	none
5	What the Stillpoint Heart unlocks	A trophy and the key to the optional Spire boss rush. Real rewards come from the author's FTB Quests	A Home Seam: a personal return Seam placed at a chosen Overworld spot
6	Pneumatic Coupler look	Same model as the original staff, renamed by the recipe (Option A)	Its own model and texture (Option B), which needs the author's art
7	Journey length	First crossing after roughly 5 to 8 hours of normal play, full completion after roughly 15 to 20	Shorter for a fast pack, longer with more favors
8	Names	The working names in section 3	The alternates in the same table

#	Decision	Locked answer	The alternative
9	Starter FTB Quests chapter	None. Only the ids (advancements, stats, tags)	A drafted chapter per act, once the mod exists
10	World size	3,000 by 3,000 blocks (radius 1,500), a fading border, and no deaths at the edge	Smaller or larger by config
11	Music	Support only. No music is shipped (section 21)	The author adds tracks later
12	Structures	Generated by script, with the optional hand polish in section 22	The author hand-builds them
13	Sound effects	Vanilla stand-ins plus synthesized files, replaced by the author later	The author supplies final files
14	Order of the Fealty work	The Fealty session starts on day one and blocks only M10	none
15	Source of truth for who builds what	The table in section 2	none

Assumptions

- **Stack:** Minecraft 1.21.1 on NeoForge. Create, Create Aeronautics (the Simulated project), Sable, GeckoLib and Fealty are required. FTB Quests and KubeJS are optional.
- **The staff:** the Pneumatic Coupler is the original creative physics staff, renamed by its recipe. Its code was read and no creative-mode check was found, so it works in survival. Confirm against the pinned version.
- **Licenses:** Skyseam depends on Aeronautics and Sable but ships none of their art or code. Sable uses the Polyform Shield license and Aeronautics' art is all rights reserved, so the author reads both licenses before publishing. This document is not legal advice.
- **References:** the four screenshots are style references. The Seam is inspired by the Dungeons rift but uses original art and sound, and no build or asset is copied.

- **Fealty:** it gains eight general features and an API bump to 1.3.0, written as a prompt in section 16, so Skyseam stays a plug-in rather than a patch.
- **No armor:** there are no armor items or trims. Rewards are tools, materials and utility.
- **Size:** the Halcyon is deliberately limited in size, as the author asked, with the Mirror Sea as its floor and a fading border.
- **Scope:** this document is the build specification. Nothing has been built yet.

Not checked yet

- Whether Sable can save a ship in one dimension and load it in another. The M0 spike answers it.
- Exact mod versions. Create Aeronautics and Sable change quickly, so pin versions when the build starts, and confirm every external id (section 1, rule 9).
- Whether the Blockbench MCP can create GeckoLib-format projects and what it can export. Claude Code records the answer in `docs/ENVIRONMENT.md` and falls back to generated JSON if needed.
- Whether Sable's native physics library loads on a headless server.
- Multiplayer and shader-pack behavior. The design is server-authoritative and avoids custom shaders, but none of it has been tested.
- The quality of generated textures, structures and sounds. Only the author can judge it.